Metal-Organic Frameworks (MOFs) - from basic research to potential applications

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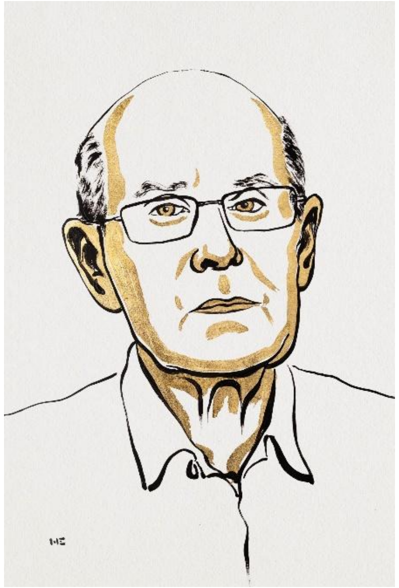
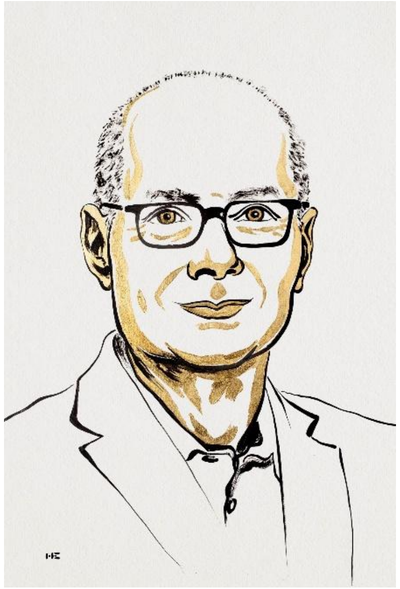
The 2025 Nobel Prize in Chemistry for Metal-Organic Frameworks (MOFs)

Susumu Kitagawa



Richard Robson



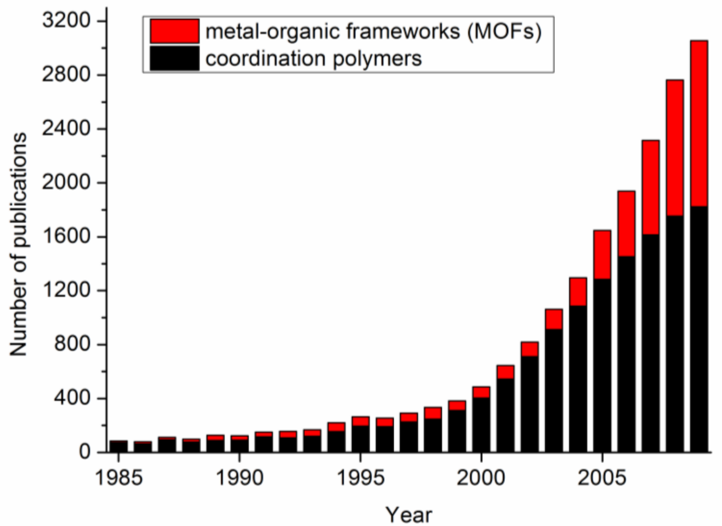
Omar M. Yaghi

"Metal–organic frameworks have enormous potential, bringing previously unforeseen opportunities for custom-made materials with new functions”

(Heiner Linke, Chair of the Nobel Committee for Chemistry)

Content

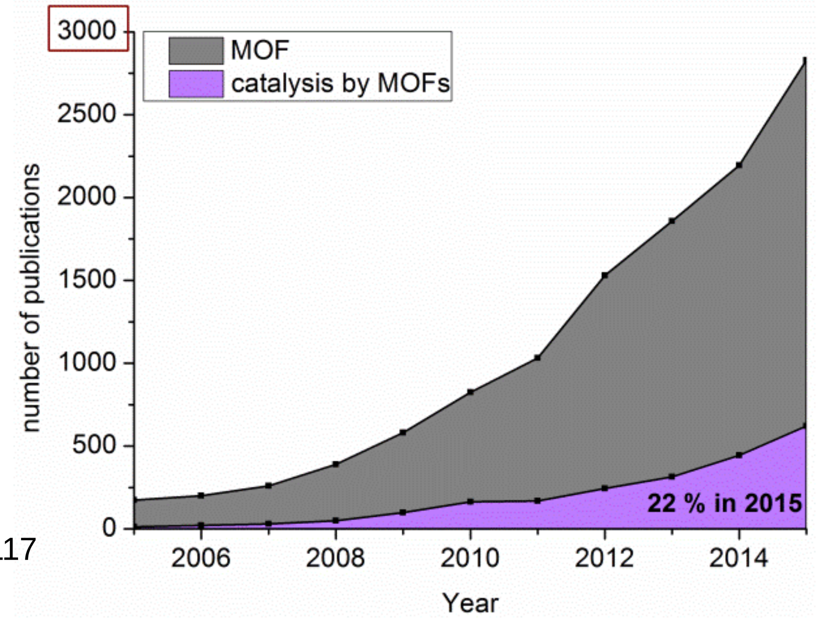
Metal-organic frameworks (MOFs):
A dynamic field



Scifinder: "metal organic framework", "MOF" or "coordination polymer"

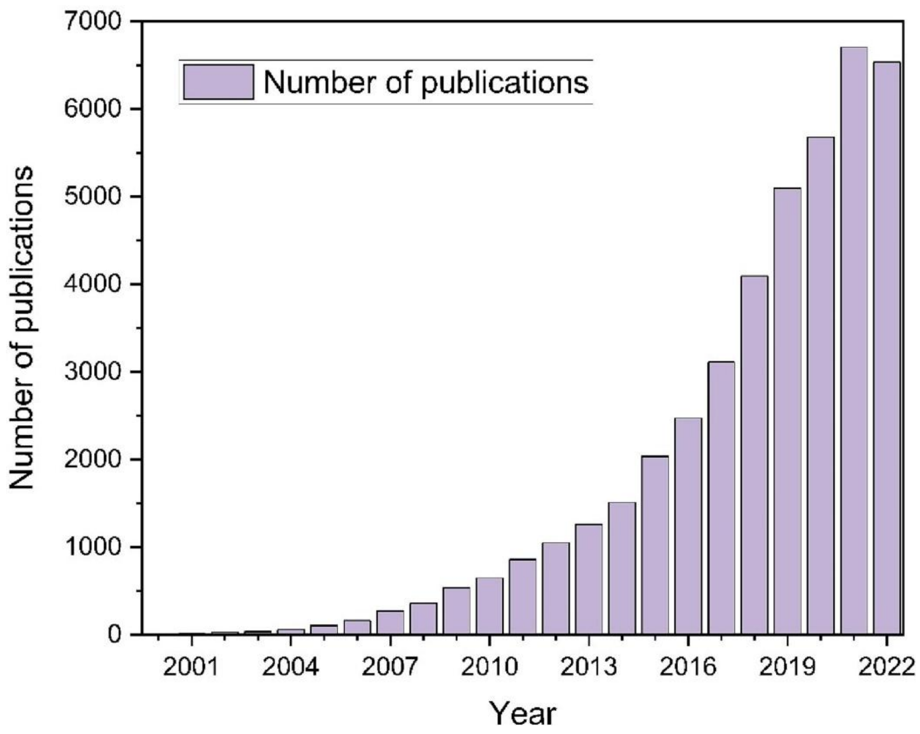
C. Janiak, J. K. Vieth
New J. Chem. **2010**, *34*, 2366-2388
DOI: 10.1039/C0NJ00275E

Scifinder: "MOF", "catalysis by MOFs"



A. Herbst, C. Janiak,
CrystEngComm **2017**, *19*, 4092-4117
DOI: 10.1039/C6CE01782G

Metal-organic frameworks (MOFs):
A dynamic field



P. Wisniewska et al., Chem. Eng. J. **2023**, 471, 144400. https://doi.org/10.1016/j.cej.2023.144400

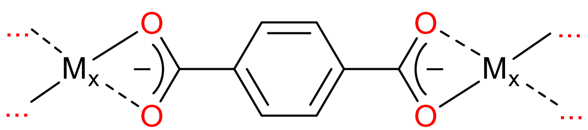
Coordination polymers / metal-organic frameworks (MOFs)

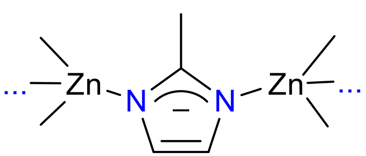
$M_{(x)}$ (or $M_{(x)}L_y$)

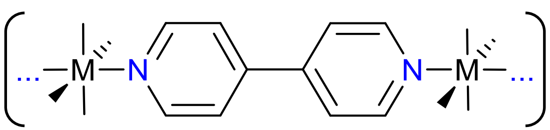
+

bridging organic ligand

Examples:

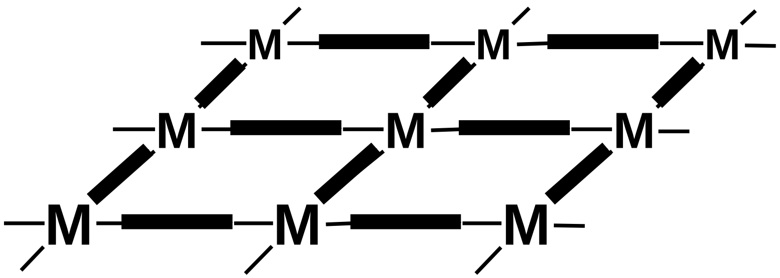




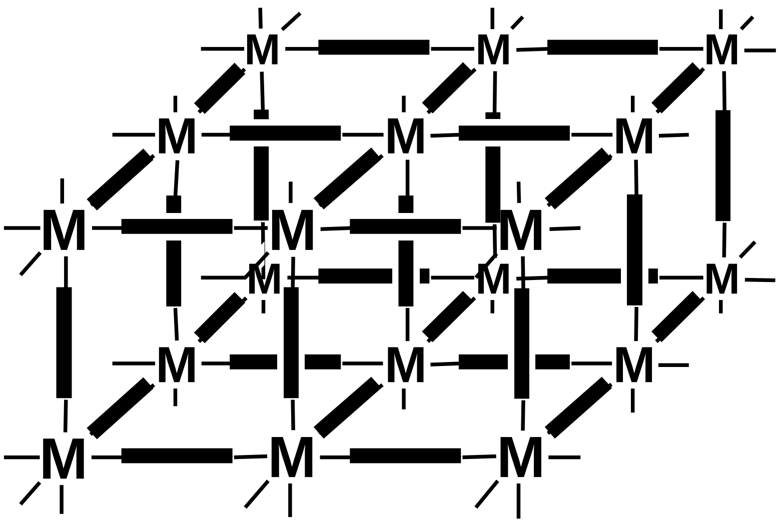


1D chains

2D nets



3D frameworks



Dalton Trans. **2003**, 2781 DOI: 10.1039/b305705b.

Metal-organic framework
- but no metal-carbon bond

until 1995:
metal-organic = organometallic = metal-carbon compounds

since introduction of "metal-organic framework", "MOF"

metal-organic = metal with organic ligand
- with O, N etc. donor atoms
- and coordinative M-O, M-N etc. bond

=> MOFs are coordination compounds

Coordination polymers / metal-organic frameworks (MOFs) – Definitions

Coordination polymer:

A coordination compound with repeating coordination entities extending in 1, 2, or 3 dimensions.

S. R. Batten, N. R. Champness, X.-M. Chen, J. G.-Martinez,S. Kitagawa, L. Öhrström, M. O'Keeffe, M. P. Suh, J. Reedijk, *CrystEngComm* **2012**, *14*, 3001–3004;
Pure Appl. Chem. **2013**, *85*, 1715-1724.

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Coordination network:
A coordination compound extending, through repeating coordination entities, in 1 dimension, but with cross-links between two or more individual chains, loops, or spiro-links, or a coordination compound extending through repeating coordination entities in 2 or 3 dimensions.

S. R. Batten, N. R. Champness, X.-M. Chen, J. G.-Martinez,S. Kitagawa, L. Öhrström, M. O'Keeffe, M. P. Suh, J. Reedijk, *CrystEngComm* **2012**, *14*, 3001–3004;
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Metal-organic frameworks:
A metal–organic framework, abbreviated to MOF, is a coordination network with **organic ligands** containing **potential voids**.

not necessarily – but most often
- 3D
- crystalline
- permanent and proven porosity
- organic bridging ligand

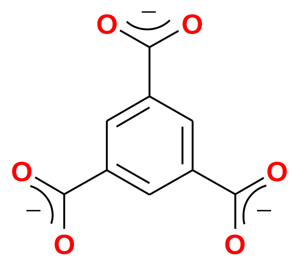
- MOFs have to be activated to achieve porosity
- crystalline nature allows for (single crystal) X-ray diffraction studies
- porosity determination by gas sorption (N_2, CO_2, H_2, Ar)

S. R. Batten, N. R. Champness, X.-M. Chen, J. G.-Martinez,S. Kitagawa, L. Öhrström, M. O'Keeffe, M. P. Suh, J. Reedijk, *CrystEngComm* **2012**, *14*, 3001–3004;
Pure Appl. Chem. **2013**, *85*, 1715-1724.

Metal-Organic Frameworks (MOFs)
– 3 examples

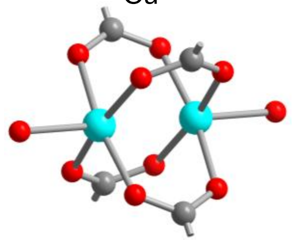
- more details later

prototypical linkers (ligands)

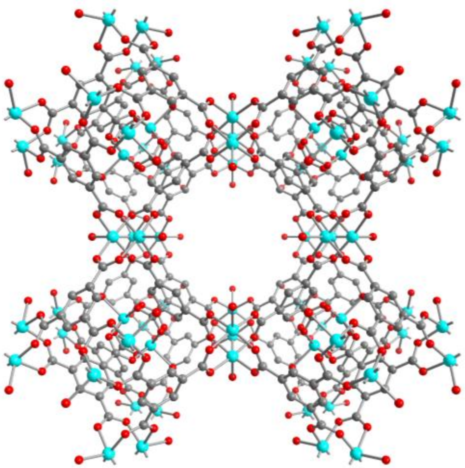


benzene-1,3,5-tricarboxylate,
trimesate, BTC
+
Cu^{2+}

metal cluster = secondary building units (SBU)



3D networks



acronyms HKUST-1

Metal-Organic Frameworks (MOFs)
– 3 examples

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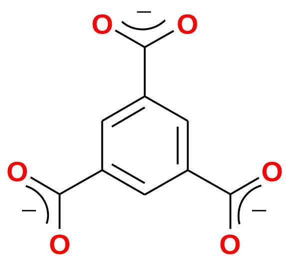
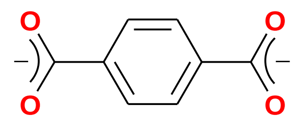
- more details later

prototypical linkers (ligands)

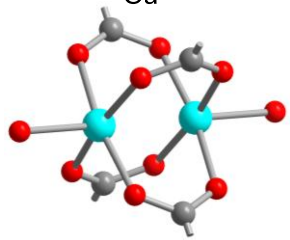


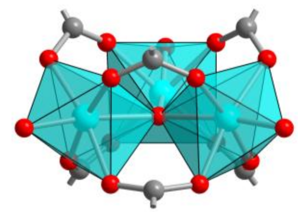
benzene-1,3,5-tricarboxylate,
trimesate, BTC
+
Cu^{2+}



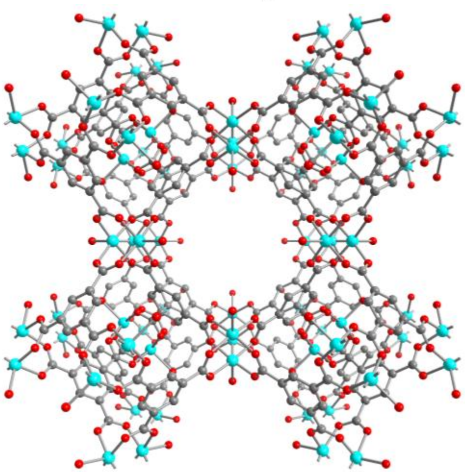
benzene-1,4-dicarboxylate,
terephthalate, BDC
+
Cr^{3+}

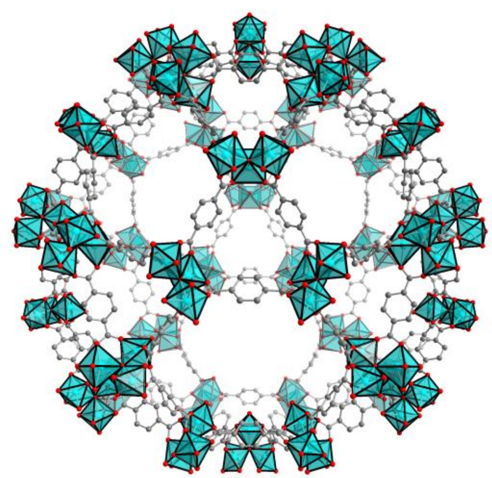
metal cluster = secondary building units (SBU)





3D networks





acronyms	HKUST-1	MIL-101

Metal-Organic Frameworks (MOFs)
– 3 examples

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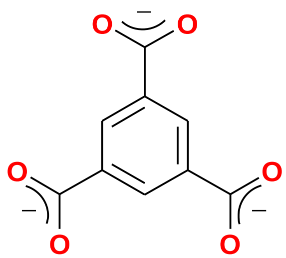
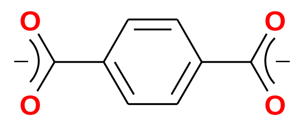
- more details later

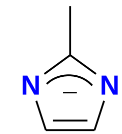
prototypical linkers (ligands)



benzene-1,3,5-tricarboxylate, trimesate, BTC
+
Cu^{2+}

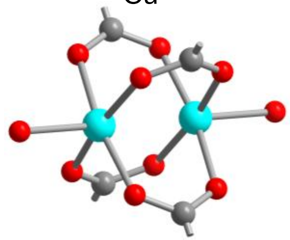


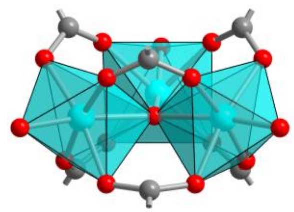
benzene-1,4-dicarboxylate, terephthalate, BDC
+
Cr^{3+}

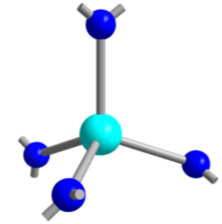


2-methyl-imidazolate
+
Zn^{2+}

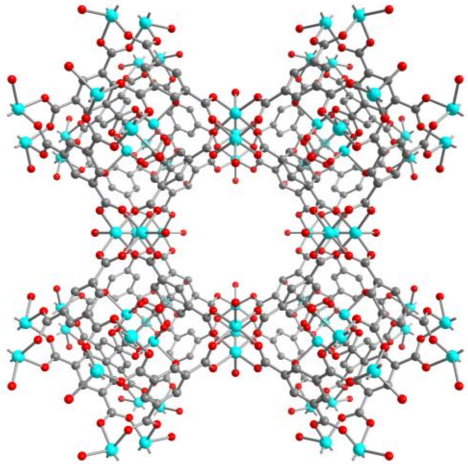
metal cluster = secondary building units (SBU)

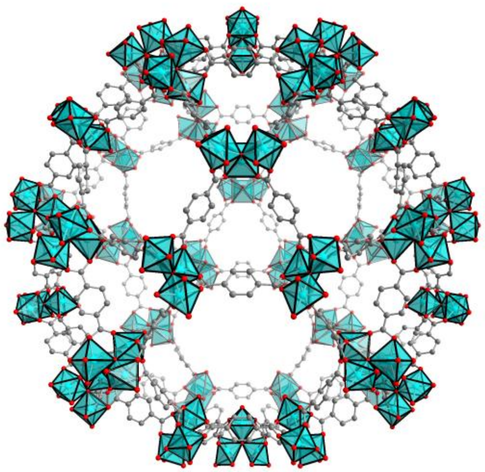


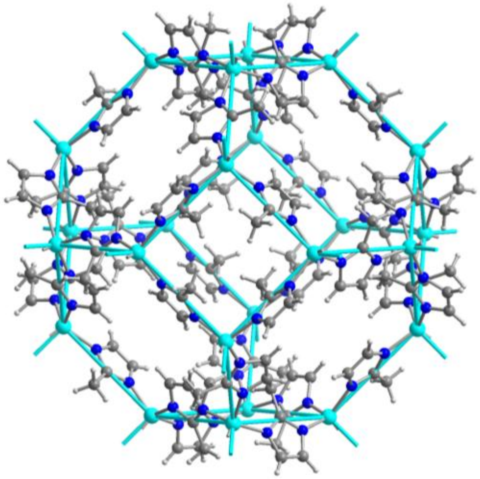




3D networks







acronyms

HKUST-1

MIL-101

ZIF-8

Prussian Blue – coordination polymer and network
- but no MOF

Prussian Blue

iron(III)-hexacyanidoferrate(II)
"$\{Fe_4[Fe(CN)_6]_3\}_\infty$"

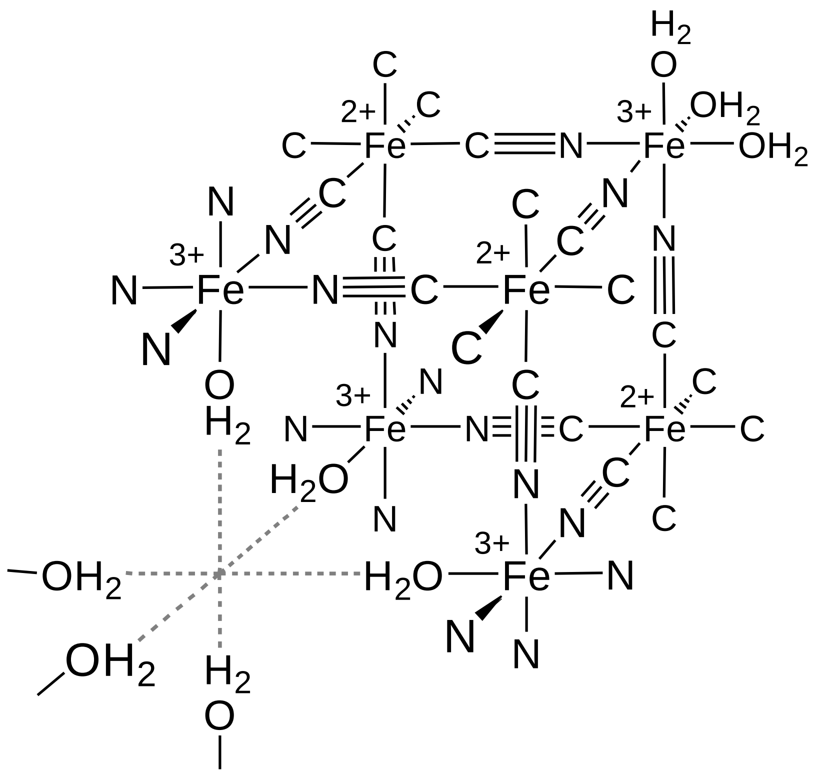
-three-dimensional (3D)
-mixed-valent
-coordination polymer;

- cyanide, CN^- bridges between octahedrally coordinated Fe-atoms;

-Fe^{2+} is C-coordinated,
-Fe^{3+} is N-coordinated by the bridging ligand;

- porous (Tl^+ or Cs^+ can be incorporated by exchange with remaining K^+ ions)

- but no organic ligands



colloidal Prussian Blue
-as medical therapeutic, oral doses of up to 20 g/day
-antidote against thallium poisoning and
-for decorporation (elimination from the body) or preventing resorption of radioactive cesium (^{137}Cs)
-trade names: Antidotum Thallii Heyl®, Radiogardase®-Cs

Important: MOFs are coordination compounds

- for the M-L coordinative bond coordination chemistry principles apply

-- spectrochemical series

I^-, Br^-, S^{2-}, ^-SCN, Cl^-, NO_3^-, F^-, OH^-, $C_2O_4^{2-}$

weak ligands

H_2O, ^-NCS, $NCCH_3$, NH_3, en

crystal/ligand-field splitting
Δ increasing →

bipy, phen, NO_2^-, ^-CN, PR_3, CO, NO^+

strong ligands

-- stability trends:

M^{2+}- < M^{3+}-complexes

((weak ligand < strong ligand))

chelate effect: $ML_6 < M(L{\cap}L)_3$

HSAB concept

Important: MOFs are coordination compounds

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HSAB concept

Metal cations – acids	Ligands – bases
hard acids	hard bases
- highly charged e.g.	e.g.
Al^{3+},	F^-
Cr^{3+}, Fe^{3+}, Co^{3+}	O-ligands
Ti^{4+}, Zr^{4+}, Hf^{4+}	amine ligands
soft acids	soft bases
Pd^{2+}, Pt^{2+}	I^-
Cu^+, Ag^+	S-ligands
Cd^{2+}	P-ligands
	C-ligands

Important: MOFs are coordination compounds

(thermodynamics)

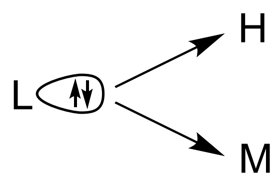
- formation/dissociation equilibrium:

$$M(solvated) + L \rightleftharpoons ML + solvent$$

H_2O is a good ligand!

- ligands are bases:

Ligand with
free electron pair
at donor atom



competition between
proton and metal ion
for ligand binding

=> pH dependency:

$$\begin{array}{ccccc} & H & & & \\ & + & & & \\ M + & L & \rightleftharpoons & ML & \\ & \uparrow\downarrow & & & \\ & HL & & & \end{array} \qquad \text{or} \qquad M + HL \rightleftharpoons ML + H$$

(charges are not shown
for simplification)

Important: MOFs are coordination compounds

(kinetics)

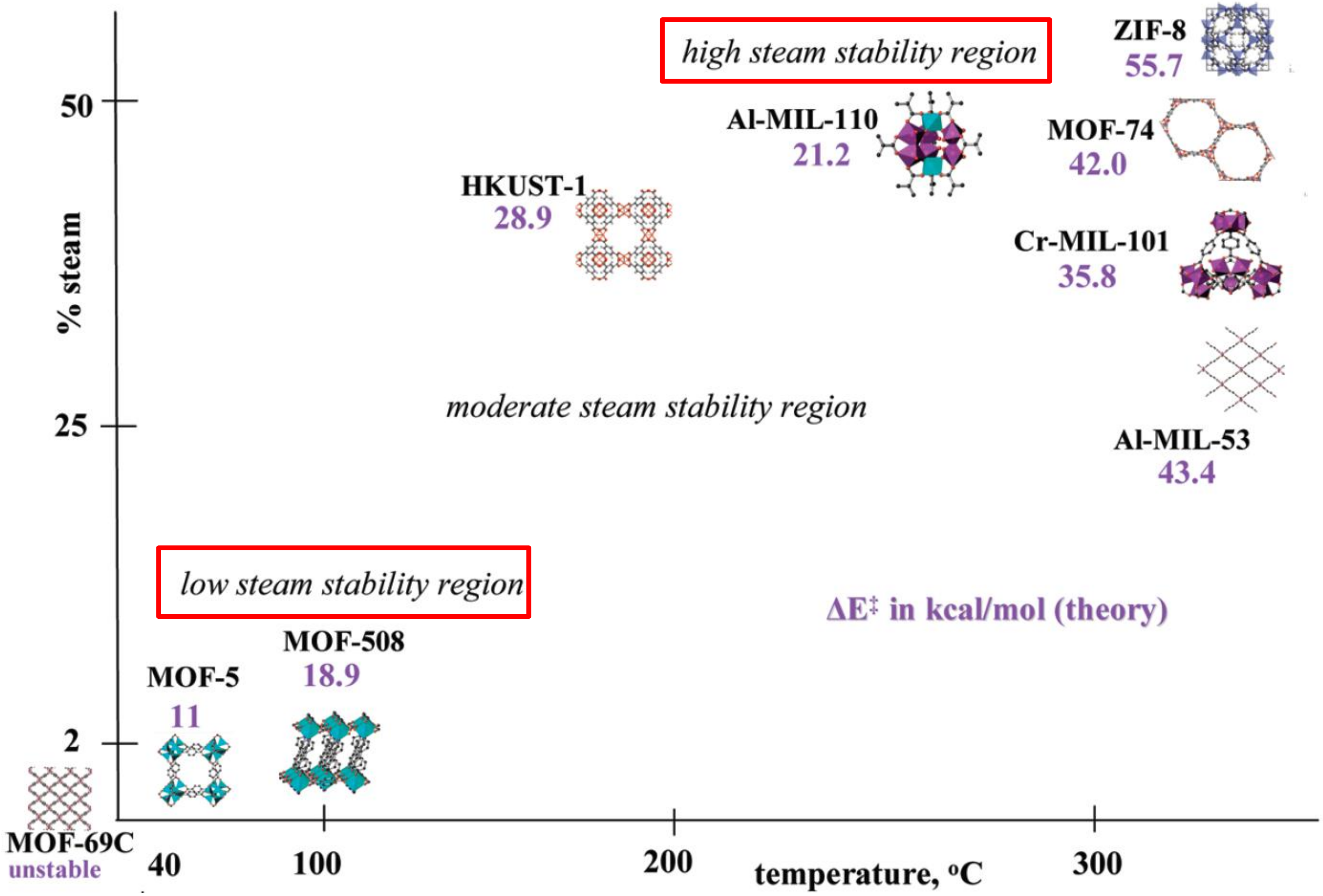
- substitution labile and inert metal-ligand bonds

inert metal ions: Cr^{3+}, Co^{3+} with d^3, d^6 (low-spin) configuration

labile metal ions: most others, e.g. Zn^{2+}

- more covalent M-L bonds are more inert than more ionic bonds
e.g. M-N more inert compared to M-O more labile

Water stability of MOFs
- an inherent problem (as water is almost everwhere)



hydrophobic!

J. J. Low, A. I. Benin, P. Jakubczak, J. F. Abrahamian, S. A. Faheem, R. R. Willis, *J. Am. Chem. Soc.* **2009**, *131*, 15834.

Water stability of MOFs
- an inherent problem (as water is almost everwhere)

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	3 days												2 months													
	pH=0		pH=4		pH=12		H_2O_2		H_2O		air		pH=0		pH=4		pH=12		H_2O_2		H_2O		air		Steam test	
	XRD	N_2	XRD	N_2	XRD	N_2	XRD	N_2	XRD	N_2	XRD	N_2	XRD	N_2	XRD	N_2	XRD	N_2	XRD	N_2	XRD	N_2	XRD	N_2	XRD	N_2
MIL-101 (Cr)																										
NH_2-MIL-101 (Al)																										
MIL-53 (Al)																										
NH_2-MIL-53		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP		NP
UiO-66																										
NH_2-UiO-66																										
UiO-67																										
CuBTC																										
ZIF-8																										

(indicated MOFs are described later)

Powder X-ray diffractogram and surface area, S_{BET}
completely
partially
destroyed

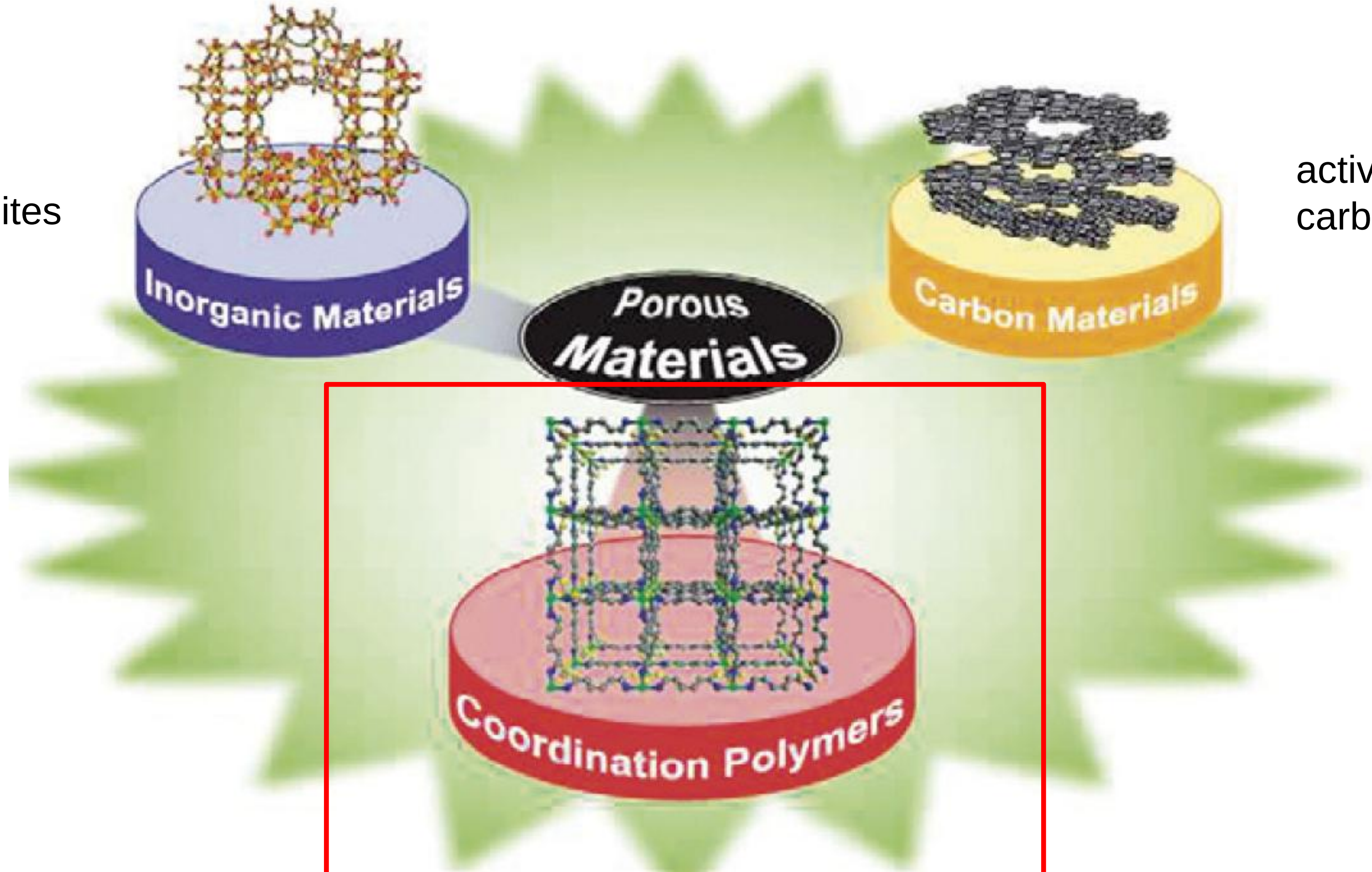
K. Leus, T. Bogaerts, J. De Decker, H. Depauw, K. Hendrickx, H. Vrielinck, V. Van Speybroeck, P. Van Der Voort, *Micropor. Mesopor. Mater.* **2016**, *226*, 110-116

Content

1. Definition
2. Comparison among porous materials
3. Brief history – the first MOFs
4. Prototypical MOFs – a closer look
5. Synthesis
6. Selected applications

Metal-Organic Frameworks (MOFs) as a new porous material class

zeolites



active carbon

metal-organic frameworks, MOFs

Kitagawa, Matsuda, *Coord. Chem. Rev.* **2007**, *251*, 2490–2509.

Porous materials

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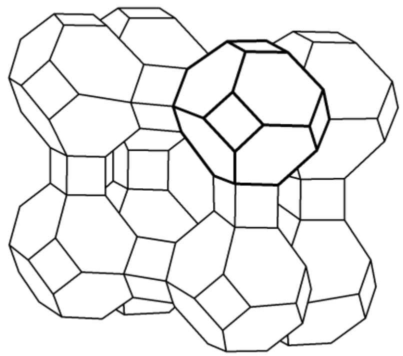
inorganic:

zeolites,
silica gel,
mesoporous silica

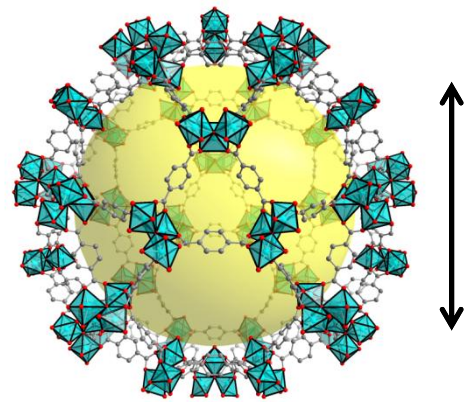


hygroscopic

applications as desiccant

inorganic-organic hybrids:

Metal-Organic Frameworks, MOFs
- defined, identical pores
- designable properties



Nano-pores

3.4 nm

organic :

active carbon

new: porous-organic polymers
e.g. Covalent-Organic Frameworks, COFs

Porous materials:
specific surface area (BET) per gram of material

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inorganic:

zeolites,
silica gel,
mesoporous silica ~1000 m^2/g

inorganic-organic hybrids:
Metal-Organic Frameworks, MOFs

~2000-4000 m^2/g

organic :

active carbon
~1000 m^2/g

What is special of MOFs? among porous materials

- higher surface area

- crystallinity => exact structure can be determined
  - -- pore system very well defined
  - -- narrow range of pore distribution

- organic linkers give great range of design options

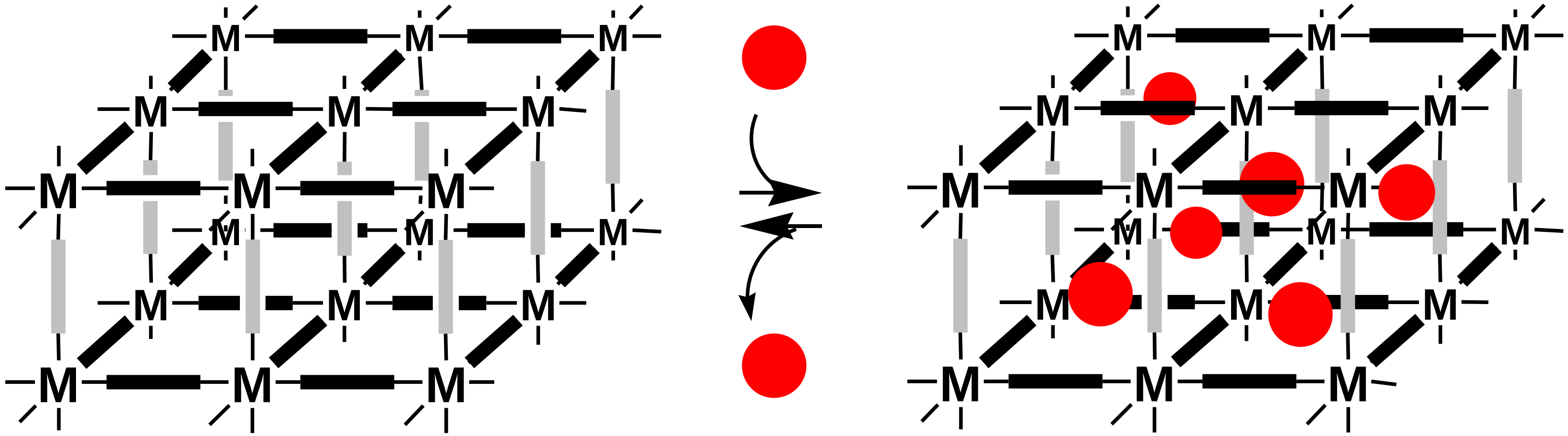
- wide variation of metal atoms and organic linkers to tune properties
  - -- hydrophilicity – hydrophobicity
  - -- pore sizes
  - -- incorporation of chirality
  - -- electronic properties

silica gel, active carbon, mesoporous silica
- amorphous
- wider range of pores

zeolites, silica gel, mesoporous silica, active carbon
- very limited range of building blocks

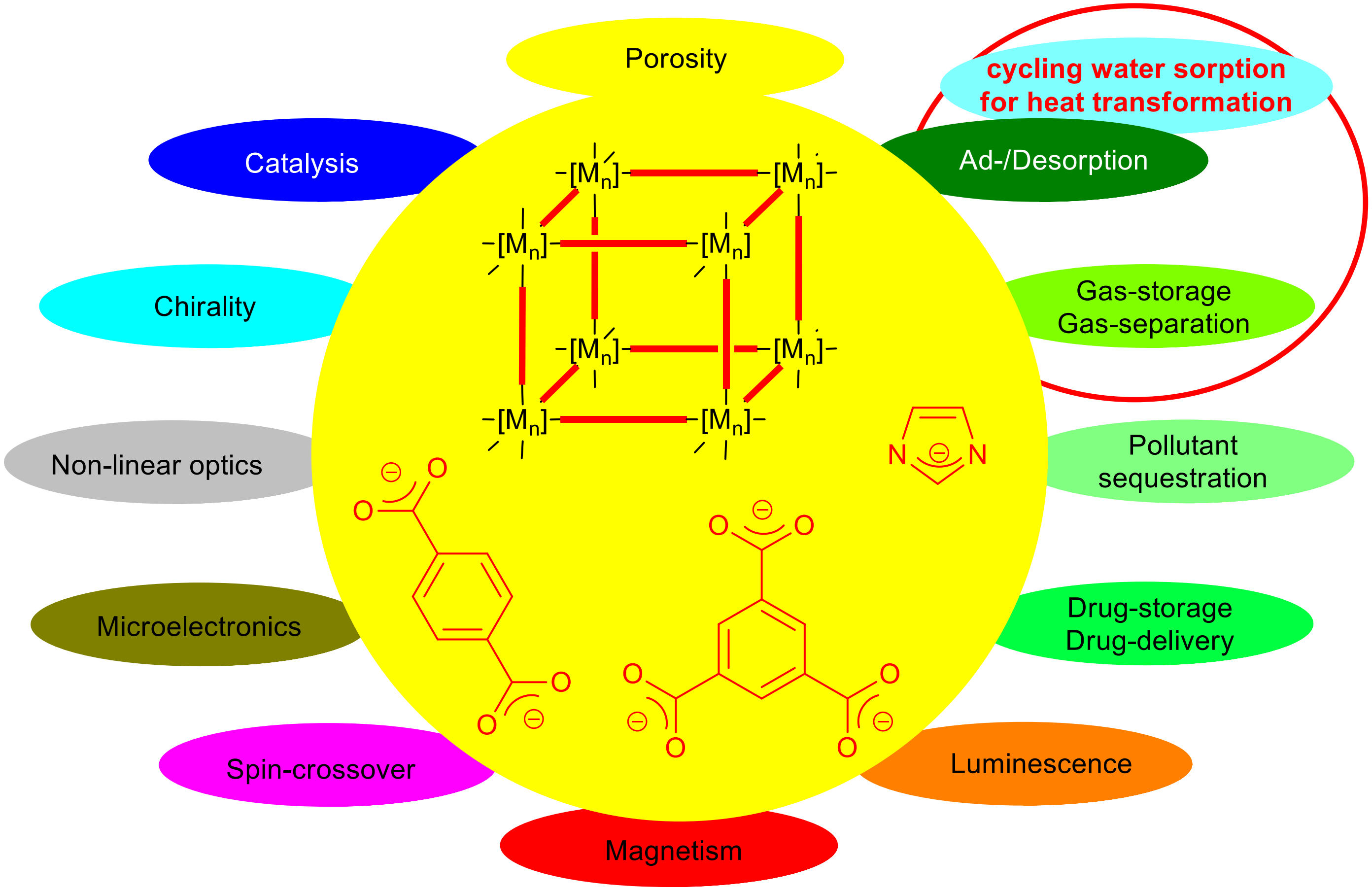
Aspect of porosity
- reversible guest exchange / storage / transformation

MOFs = "organic zeolite analogs"



Applications of zeolites, silica gel,
mesoporous silica, active carbon:
– ad/desorption
– ion exchange (e.g. water softening in detergents)
– separation (chromatography)
– drying agent ("molecular sieve")
– selective catalysis (petrochemical refining, cracking)
etc.

Metal-Organic Frameworks (MOFs):
Application-oriented properties



Reviews: *New J. Chem.* **2010**, *34*, 2366 DOI: 10.1039/C0NJ00275E.
Dalton Trans. **2003**, 2781 DOI: 10.1039/b305705b.

Content

1. Definition
2. Comparison among porous materials
3. Brief history – the first MOFs
4. Prototypical MOFs – a closer look
5. Synthesis and analysis
6. Selected applications

MOF history

- starting ~1990
- an excellent example of interdisciplinary research

-- first molecule-based field of **crystal engineering / coordination chemistry**

-- later zeolite chemists, solid-state chemists, material scientists

- coordination polymers preceded MOFs:
-- term **coordination polymers** coined and area first reviewed in **1964**.

Coordination polymers – first(?) notion

CHAPTER 1

Coordination Polymers

JOHN C. BAILAR, JR.

University of Illinois, Urbana, Illinois

CONTENTS

from: Preparative Inorganic Reactions, ed. W. L. Jolly, Interscience, New York, 1964, vol. 1, pp. 1-25.

MOF history

-- term **coordination polymers** coined and area first reviewed in 1964.

-- **interest in porous coordination polymers came much later**,

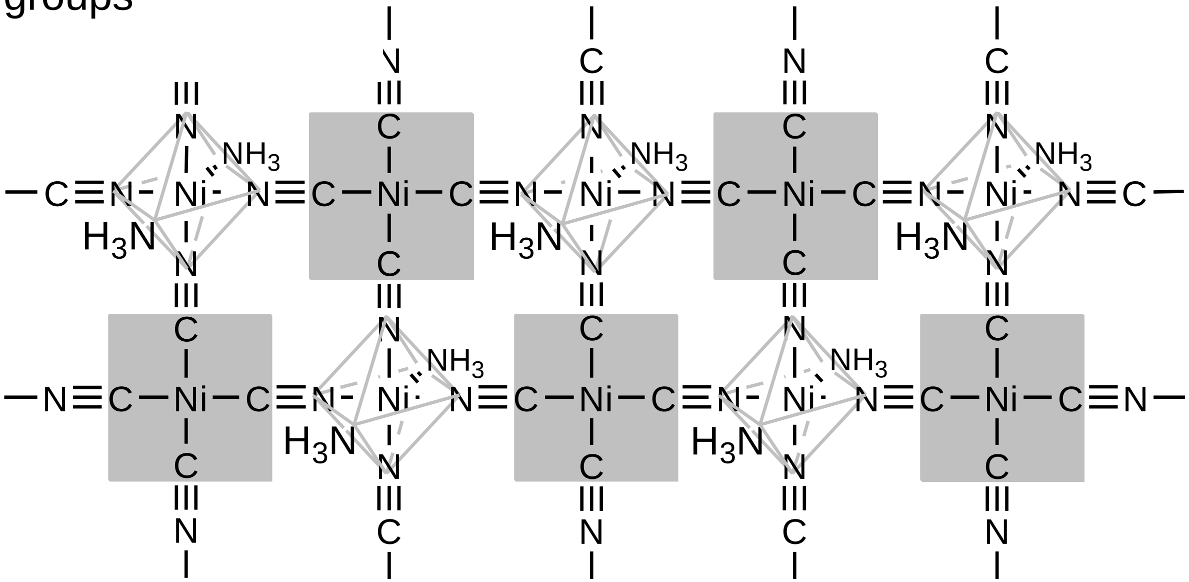
-- although some coordination complexes such as

--- Prussian blue compounds (see above) and
--- Hofmann clathrates

had been known to show reversible sorption properties.

Hofmann clathrates

2D cyanido-metal framework, constructed from
- alternating square-planar $Ni^{II}(CN)_4$-units and
- octahedral $Ni^{II}(NC)_4(NH_3)_2$-groups



- guest molecules are not shown
-- guests can be organic molecules, especially aromatics
-- reversible guest removal can be possible

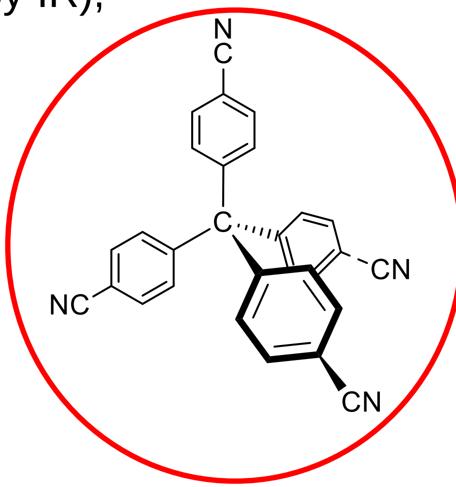
- instead of Ni in the octahedral position also Cd can be used.
- when bridging amine ligands, e.g., 4,4'-bipyridine are used in place of NH_3-groups then the 2D layers are vertically interconnected to a 3D network

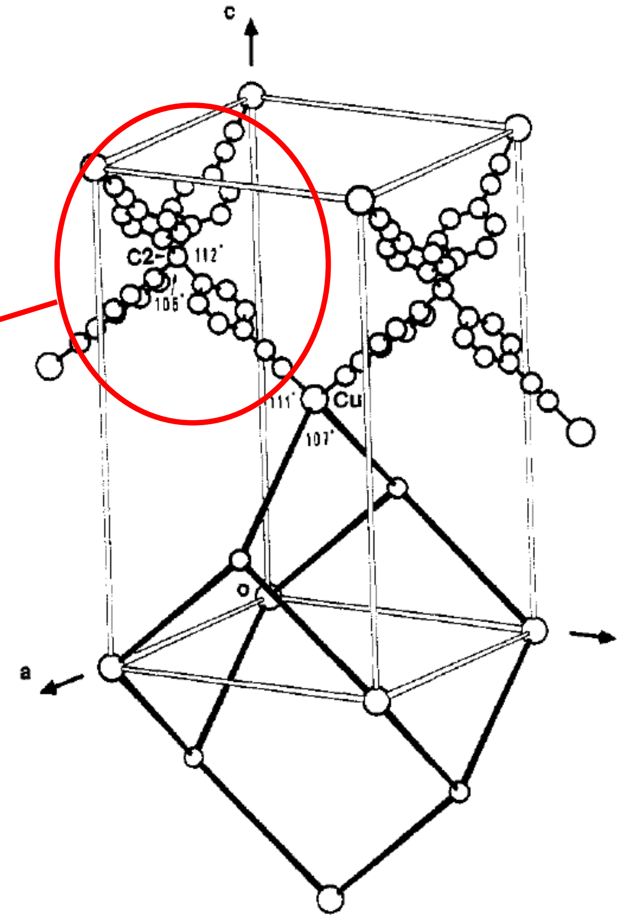
MOF history

interest in **porous coordination polymers** and MOFs started only **around 1990**.
-work by **Hoskins and Robson** (Univ. Melbourne) set the basis for the future of MOFs.

$Cu[L]BF_4 \cdot x\ C_6H_5NO_2$
- anion exchange with PF_6^- (by IR),
- diamond-related framework
- large adamantane-like cavities of ~1500 Å³

tetracyanotetraphenylmethane



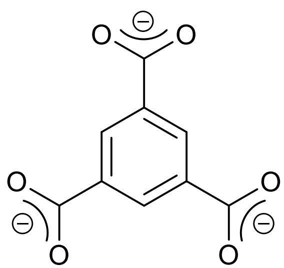


Hoskins and Robson envisioned
-large range of **crystalline, microporous, stable solids**,
-possibly using structure-directing agents,
-with **ion-exchange, gas sorption, or catalytic properties**,
-**postsynthetic modifications**

B. F. Hoskins, R. Robson, *J. Am. Chem. Soc.* **1990**, *112*, 1546-1554.

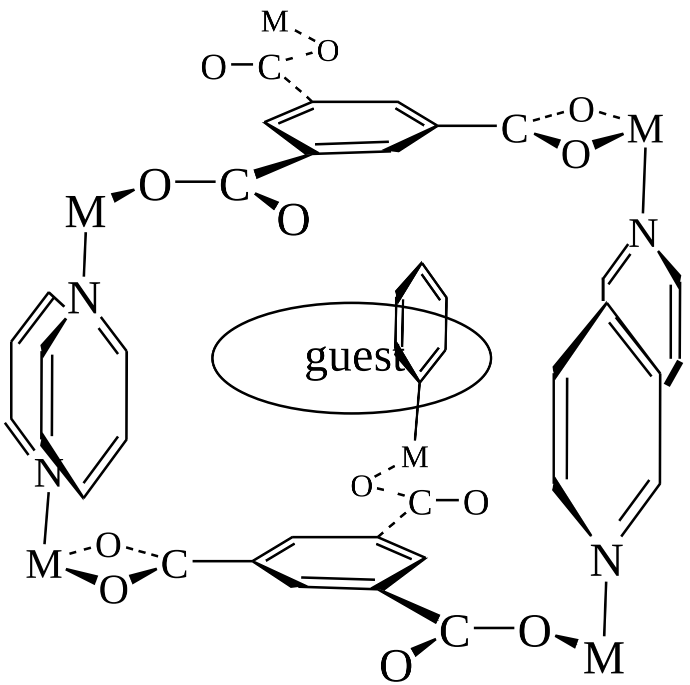
MOF history

The **term MOF was popularized by Yaghi** et al.
around 1995
for a layered Co-trimesate that showed reversible sorption properties.



btc^{3-}

trimesate, BTC (btc)
= benzene-1,3,5-tricarboxylate



O. M. Yaghi, G. Li, H. Li, *Nature* **1995**, *378*, 703-706.

MOF history

Framework integrity in the absence of guest molecules?

-layer structure, with the pyridine ligands functioning as spacer units between the layers.

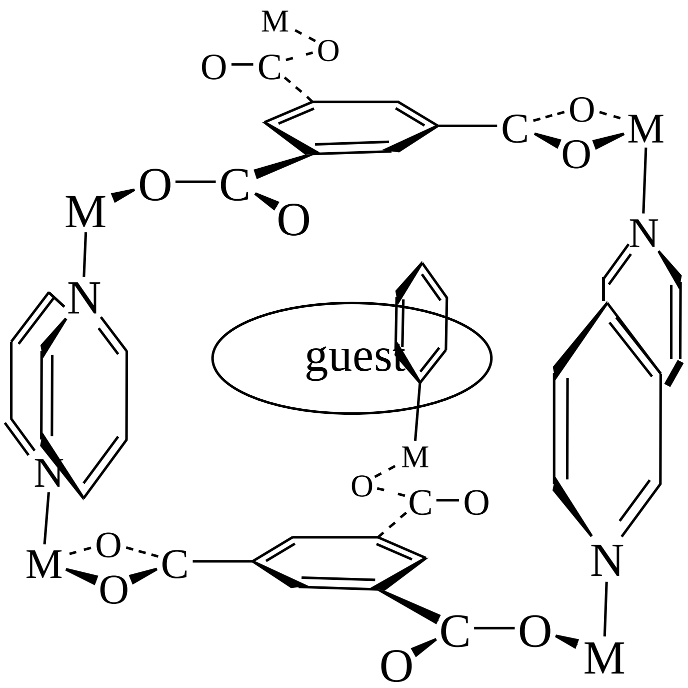
-7x10 Å rectangular channels,

-aromatic guest can reversibly be removed.

- single crystals retain their morphology and crystallinity upon loss of the guests according to optical microscopy.

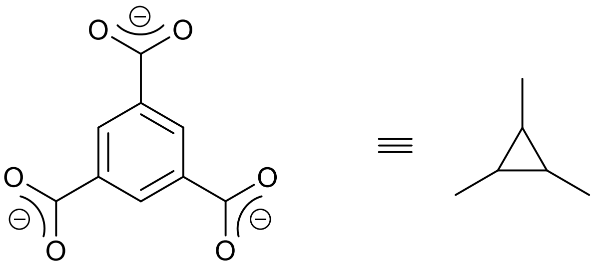
- from mixtures of components only the aromatic molecules such as benzene, nitrobenzene, cyano- or chlorobenzene are selectively absorbed (by IR).

-crystal lattice is thermally stable up to 350 °C.



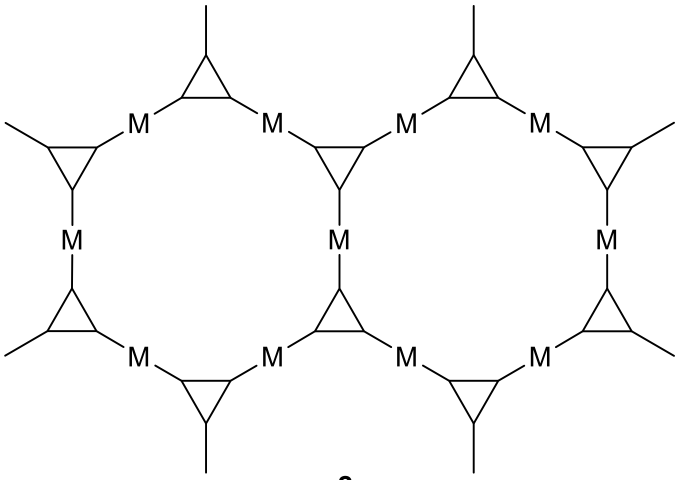
O. M. Yaghi, G. Li and H. Li, *Nature*, 1995, **378**, 703-706.

MOF history



1, btc^{3-}

M = Co^{2+}, Ni^{2+} and Zn^{2+}



2

Rigorous proof of framework integrity in the absence of guest molecules still needed to be developed.

hydrated solid of formula $M_3(btc)_2 \cdot 12H_2O$

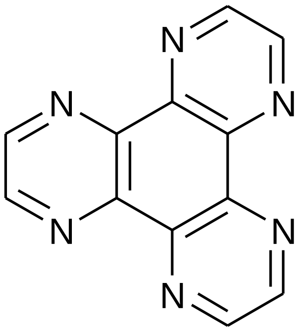
-2D network structure with 1D channels

-reversibly and repeatedly binds water,
-inclusion of ammonia was demonstrated,
-larger molecules and those without a reactive lone pair did not enter

-retention of the 2D metal-ligand assembly with 1D channels (**2**) *was proposed* for the porous solid

O. M. Yaghi, H. Li, T. L. Groy, *J. Am. Chem. Soc.*, **1996**, *118*, 9096-9101.

MOF history



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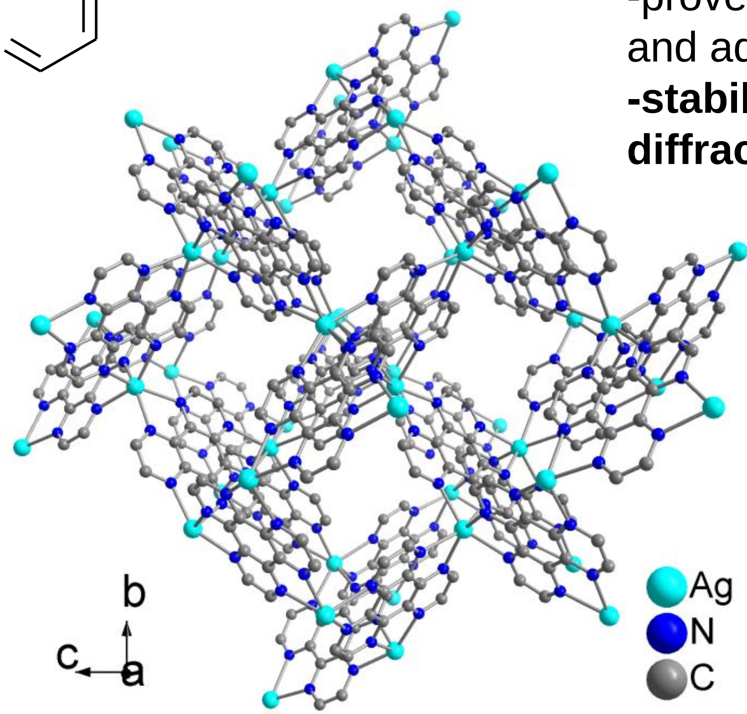


Rigorous proof of framework integrity!

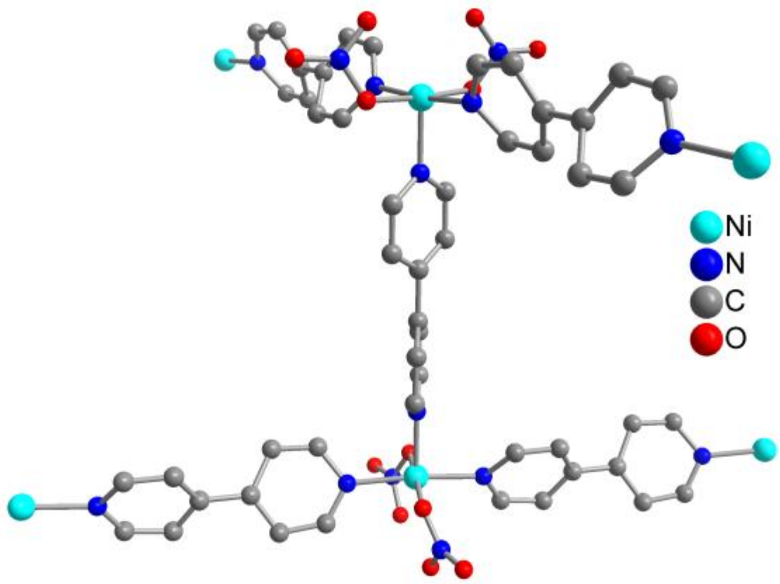
three-dimensional 10,3-a network from Ag(I) and ligand 1,4,5,8,9,12-hexaazatriphenylene (hat, **4**)
-proved stable over two nitromethane guest de- and adsorption cycles
-stability evidenced by single-crystal X-ray diffraction over four generations.

B. F. Abrahams, P. A. Jackson, R. R. Robson, *Angew. Chem. Int. Ed. Engl.*, **1998**, *37*, 2656-2659.

MOF history

Rigorous proof of framework integrity!

Kepert and Rosseinsky proved the rigidity of the two-dimensional (2D) coordination polymer network of $[Ni_2(4,4'\text{-bipy})_3(NO_3)_4]$ (bipy = bipyridine) quantitatively by single-crystal X-ray structures of both ethanol-loaded and ethanol-free forms.



Extended building block of the 2D network $[Ni_2(4,4'\text{-bipy})_3(NO_3)_4]$ formed by T-shaped coordination of 4,4'-bipyridine to Ni^{2+} with nitrate ligands.

C. J. Kepert and M. J. Rosseinsky, *Chem. Commun.*, **1999**, 375-376.

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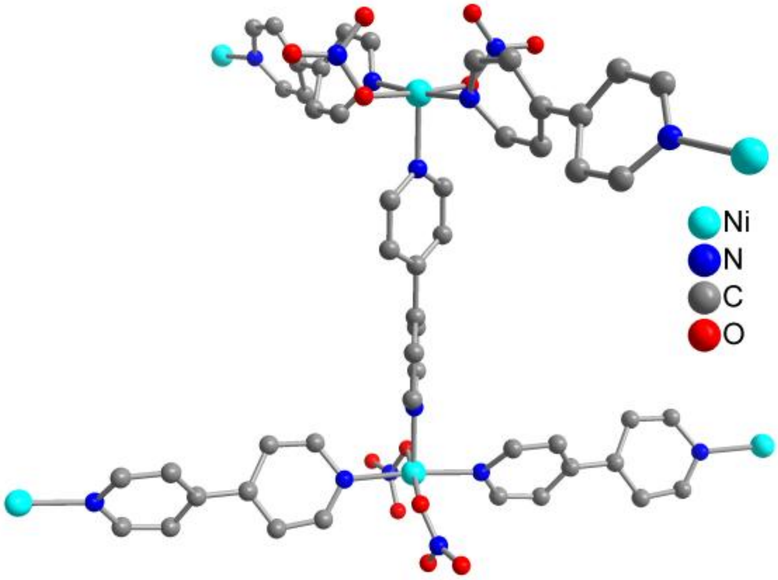

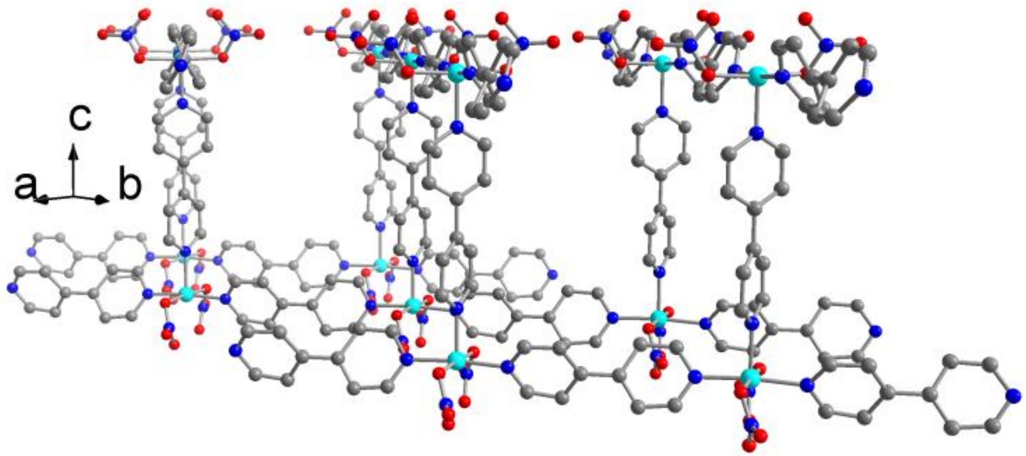

Bilayer with the 'tongue-and-groove' motif.

Extended building block of the 2D network $[Ni_2(4,4'\text{-bipy})_3(NO_3)_4]$ formed by T-shaped coordination of 4,4'-bipyridine to Ni^{2+} with nitrate ligands.

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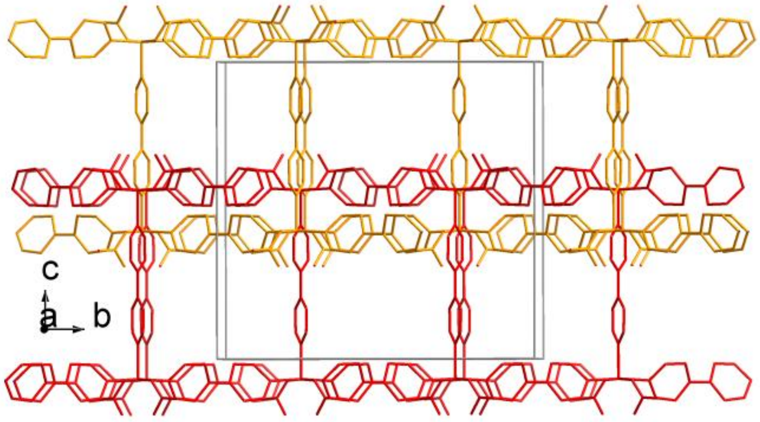

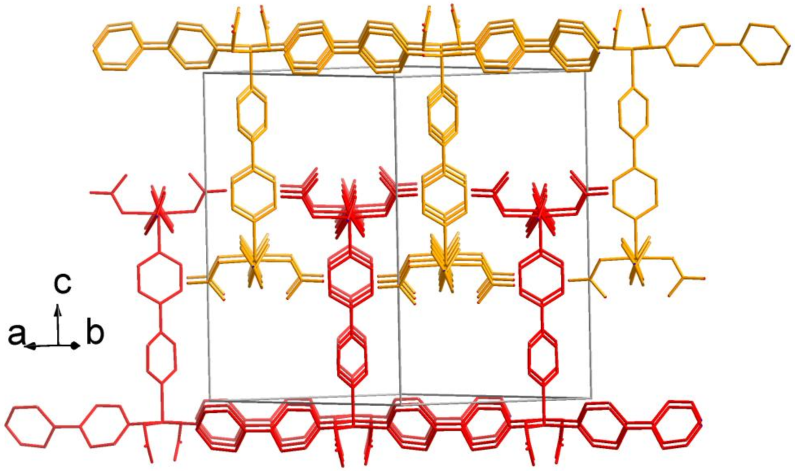

Interdigitation of two adjacent bilayers which are differentiated by color (in wire mode and two different viewing directions).

C. J. Kepert and M. J. Rosseinsky, *Chem. Commun.*, **1999**, 375-376.

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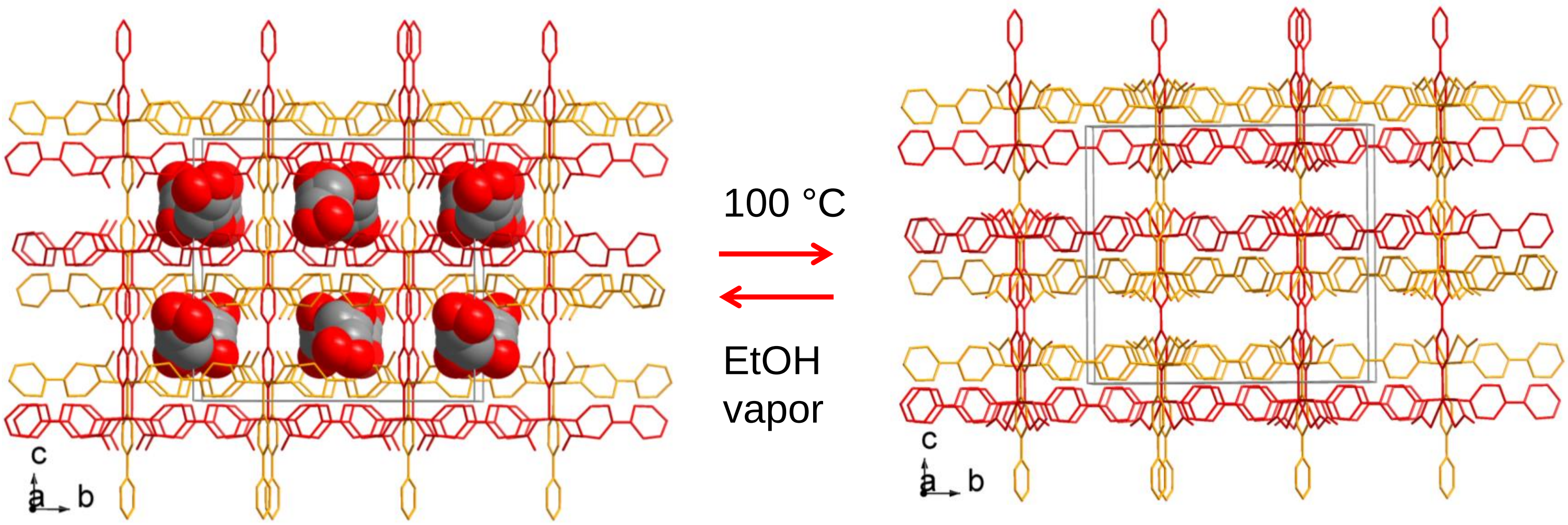

Ethanol-solvated and -emptied form. Channels with cavities run along the crystallographic *a* direction and occupy ~20% of the unit cell volume in both forms. Disordered ethanol molecules are shown in space-filling mode.

C. J. Kepert and M. J. Rosseinsky, *Chem. Commun.*, **1999**, 375-376.

MOF history

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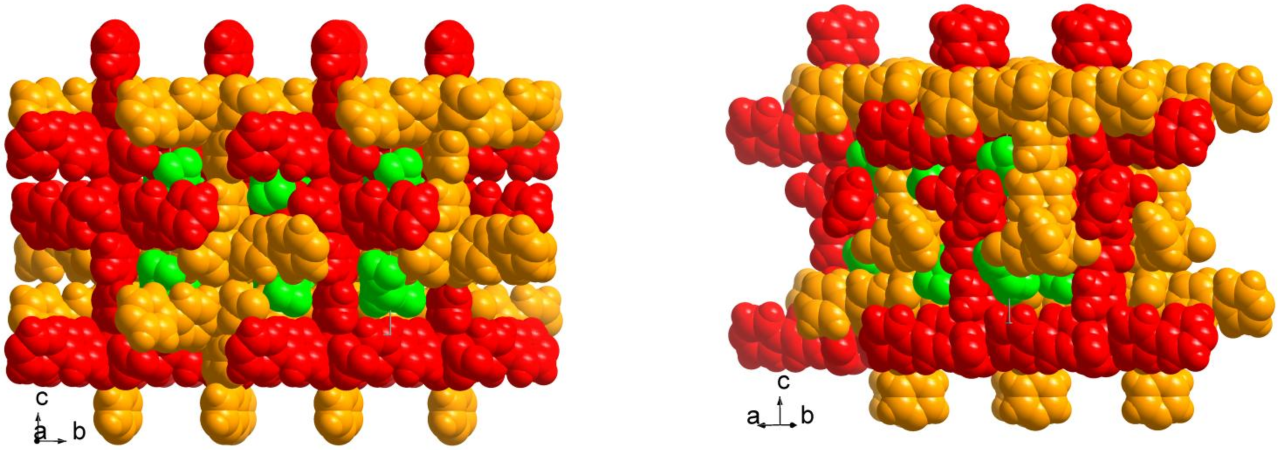


A closer examination of the structure in space-filling mode reveals that the [Ni_2(4,4'-bipy)$_3$(NO_3)$_4$] network has only very small channels with dimension of ca. 1.1 Å x 1.1 Å. Therefore, the guest molecules are rather located in cavities ("zero-dimensional closed space"). => Framework flexibility for ad/desorption necessary.

Network of the ethanol-solvated structure of [Ni_2(4,4'-bipy)$_3$(NO_3)$_4$] in two viewing directions; the ethanol solvent molecules are depicted in green color.

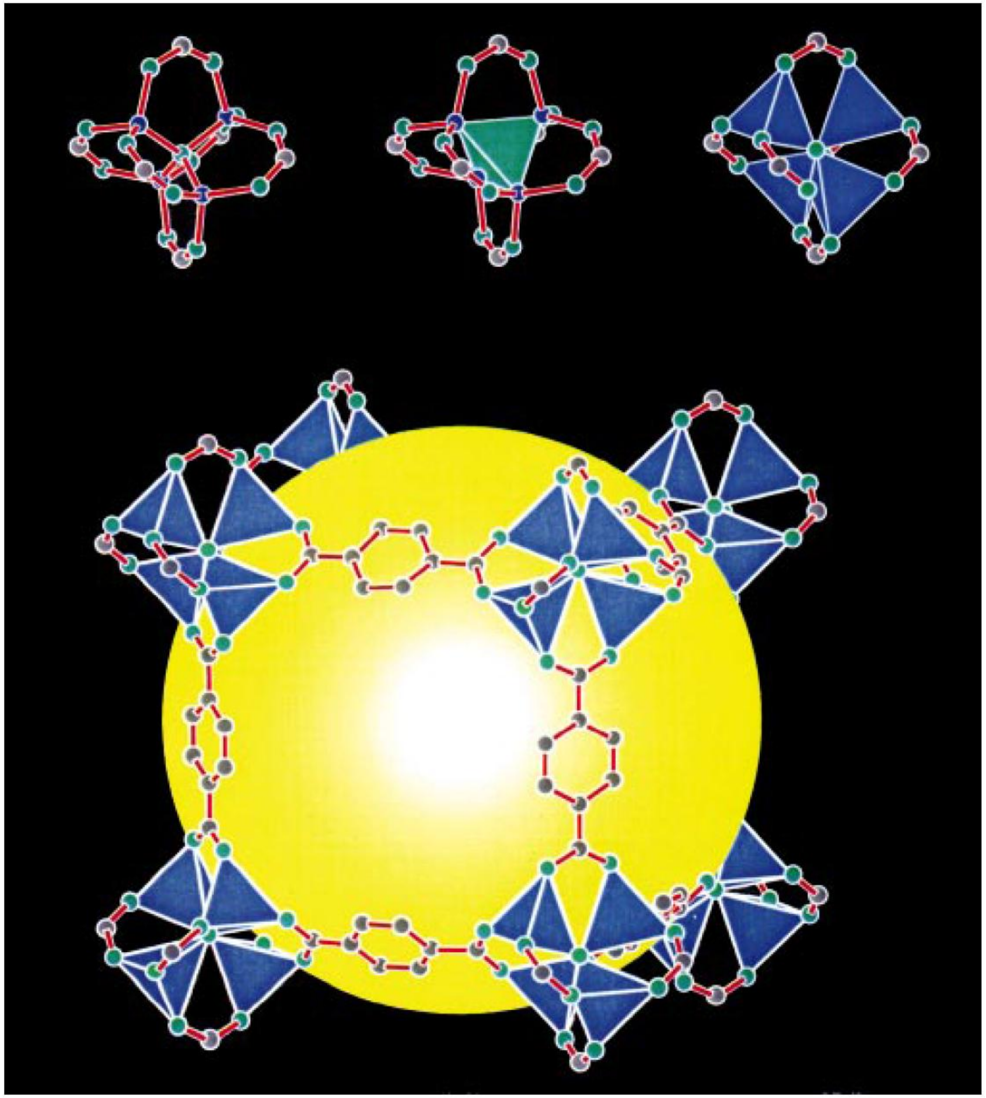
C. J. Kepert and M. J. Rosseinsky, *Chem. Commun.*, 1999, 375-376.

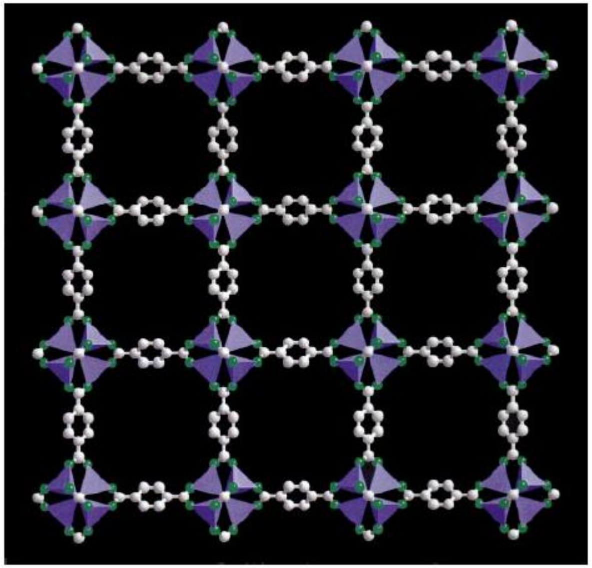
MOF history

In **1997**, a 3D MOF was reported by Kitagawa et al. that exhibited gas sorption properties at room temperature.

In **1999**, the syntheses of the prototypical **MOF-5** and **HKUST-1** were reported that in the following years and up to now are among the most studied MOFs.

MOF example:
3D-[$Zn_4(\mu_4\text{-}O)(bdc)_3$] (also called MOF-5 or IRMOF-1)





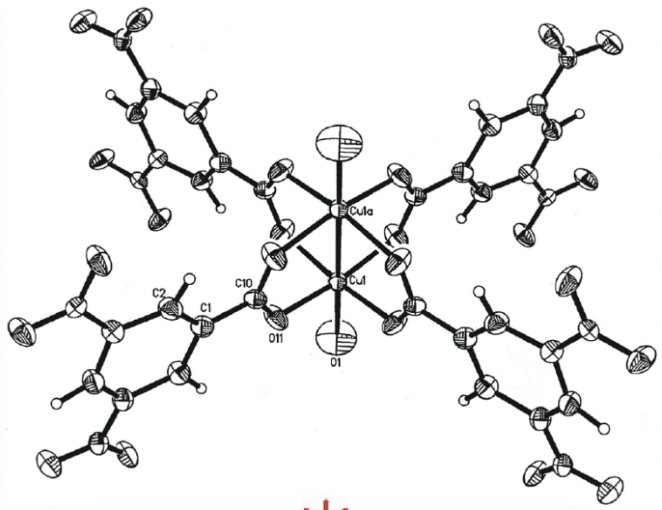
Langmuir surface area
~2900 m^2/g

H. Li, M. Eddaoudi, M. O'Keeffe, O. M. Yaghi, *Nature* **1999**, 402, 276-279.

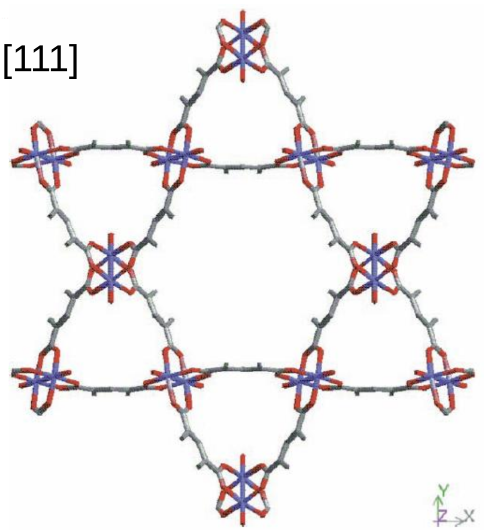
MOF example:
3D-[$Cu_3(btc)_2(H_2O)_3$] (also called HKUST-1 or Cu-btc)

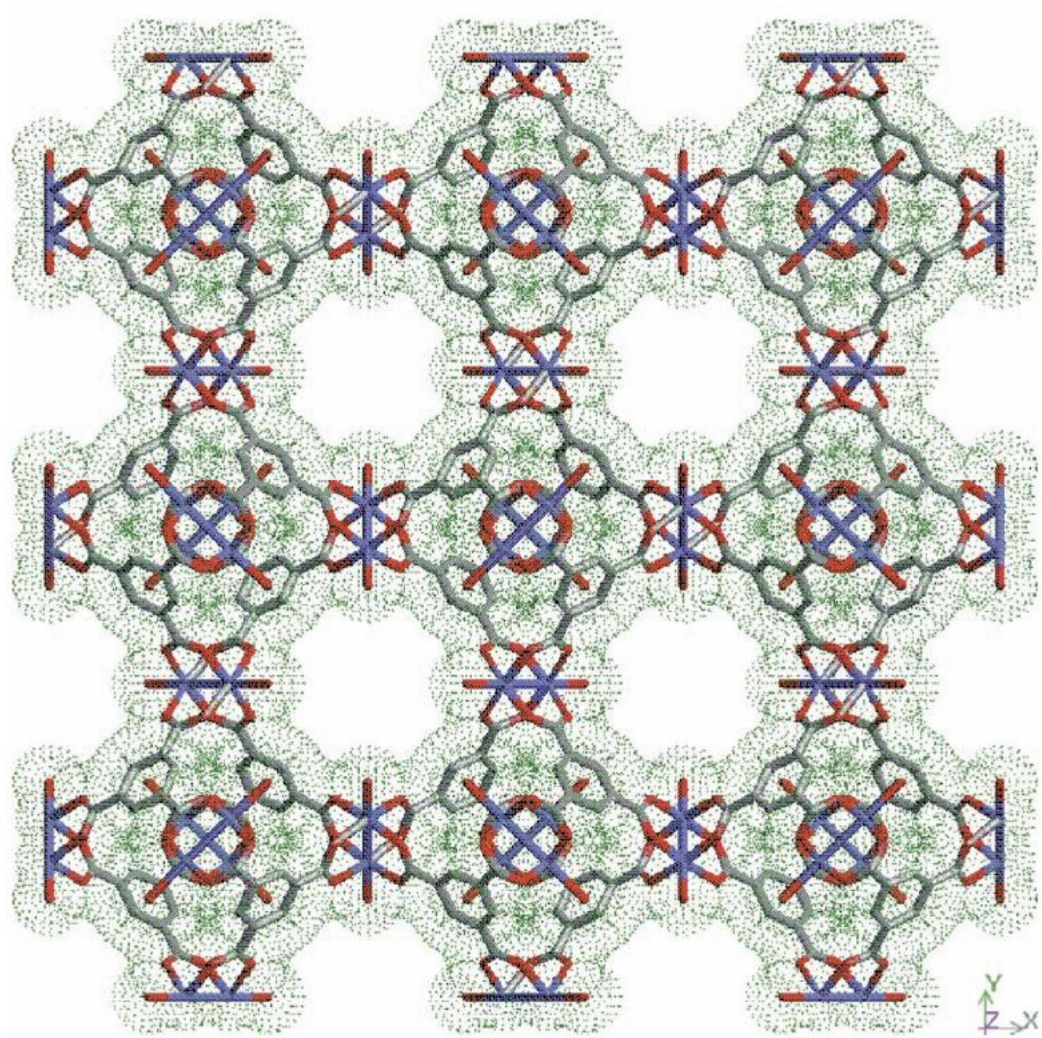
HKUST = Hongkong University of Science and Technology

along [100]



along [111]





S. S.-Y. Chui, S. M.-F. Lo, J. P. H. Charmant, A. G. Orpen, I. D. Williams, *Science* **1999**, 283, 1148-1150.

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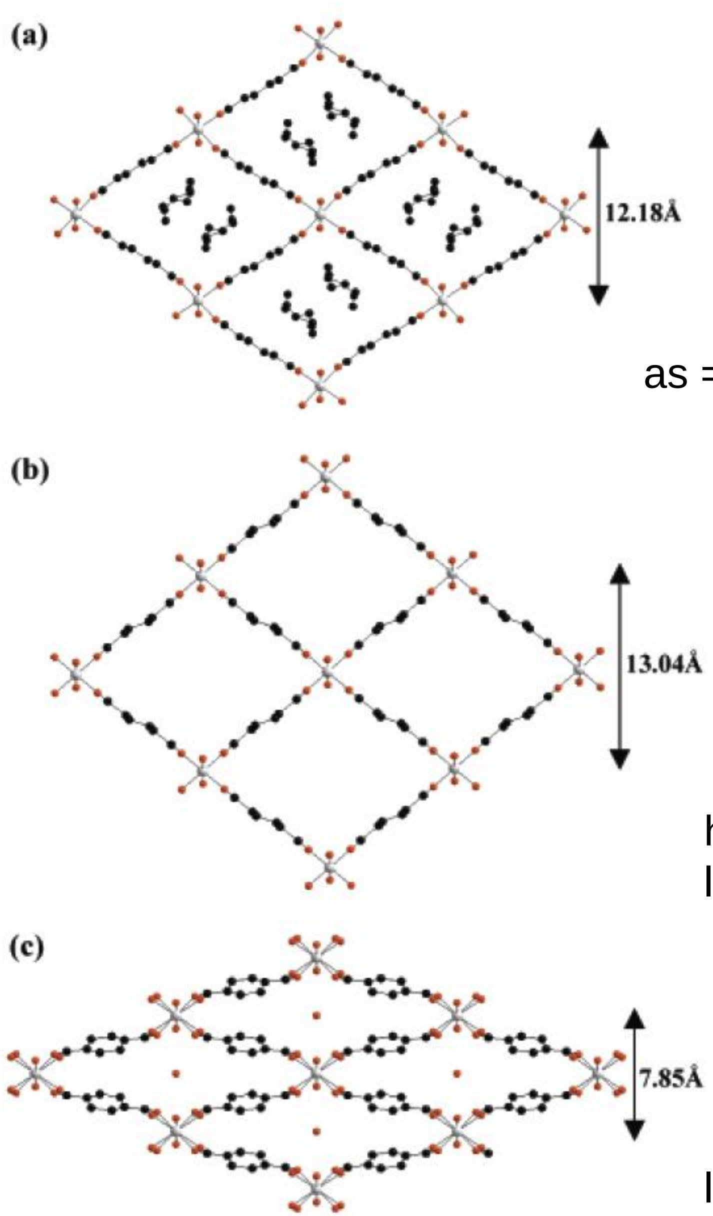
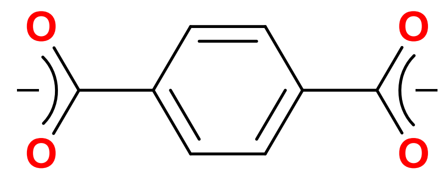
MILs
3D-[Cr(μ_4-bdc)(μ-OH)]



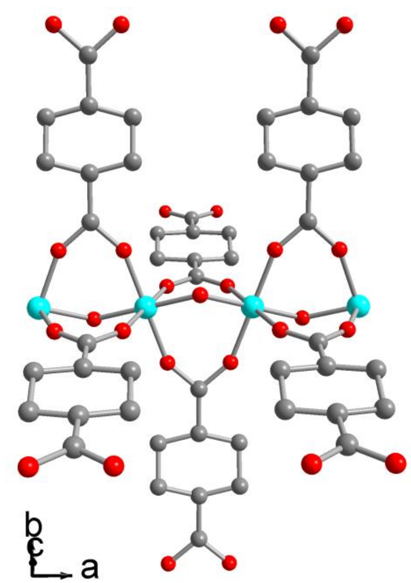
as = as synthesized

high temp.,
large pore

low temp.



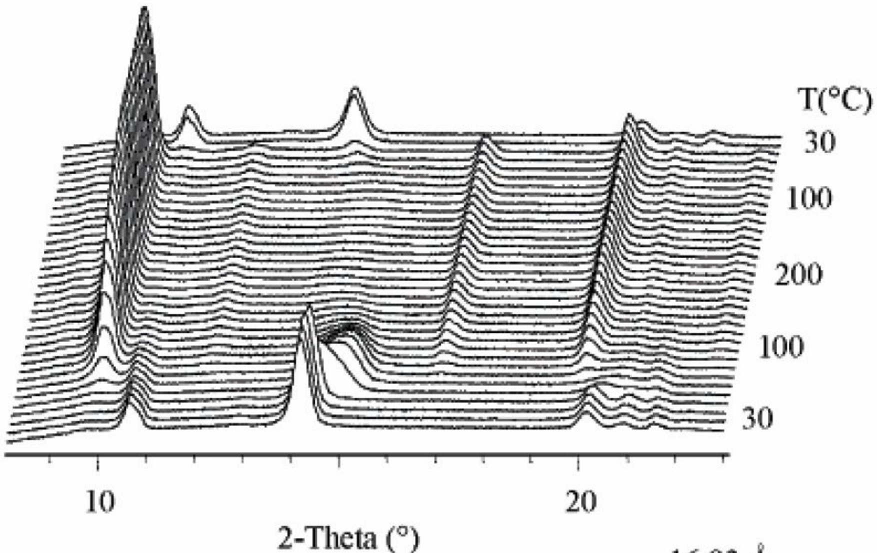
benzene-1,4-dicarboxylate,
terephthalate, bdc

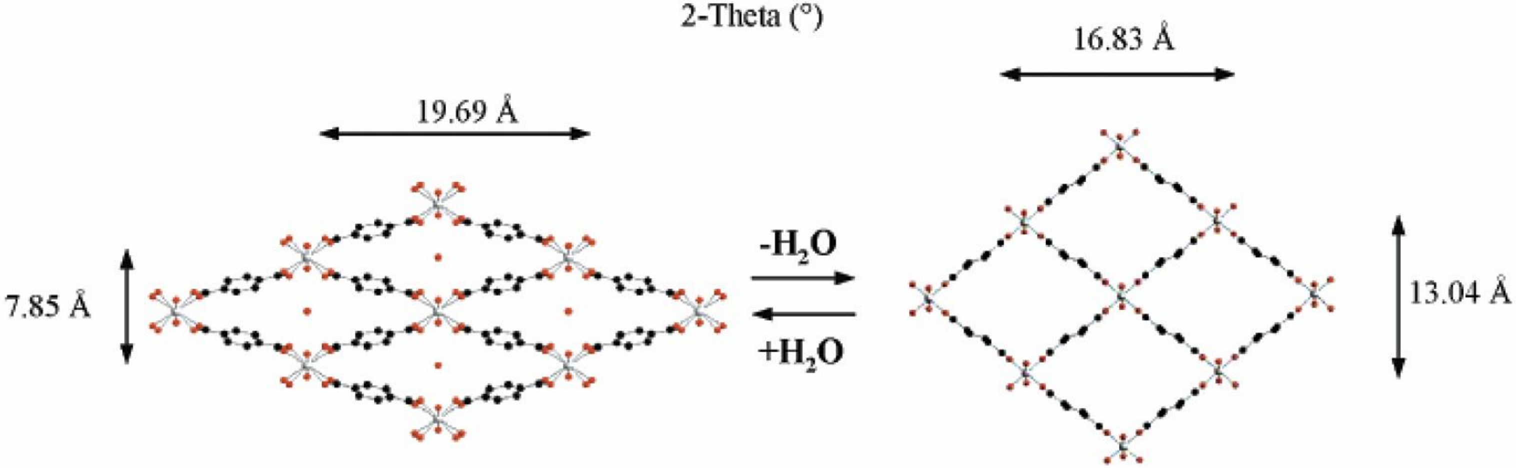


C. Serre, F. Millange, C. Thouvenot, M. Nogues, G. Marsolier, D. Louer, G. Férey, *J. Am. Chem. Soc*. **2002**, *124*, 13519-13526.

MIL-53: Prototypical breathing MOF
3D-[Cr(μ_4-bdc)(μ-OH)]

powder X-ray diffractograms (PXRD):





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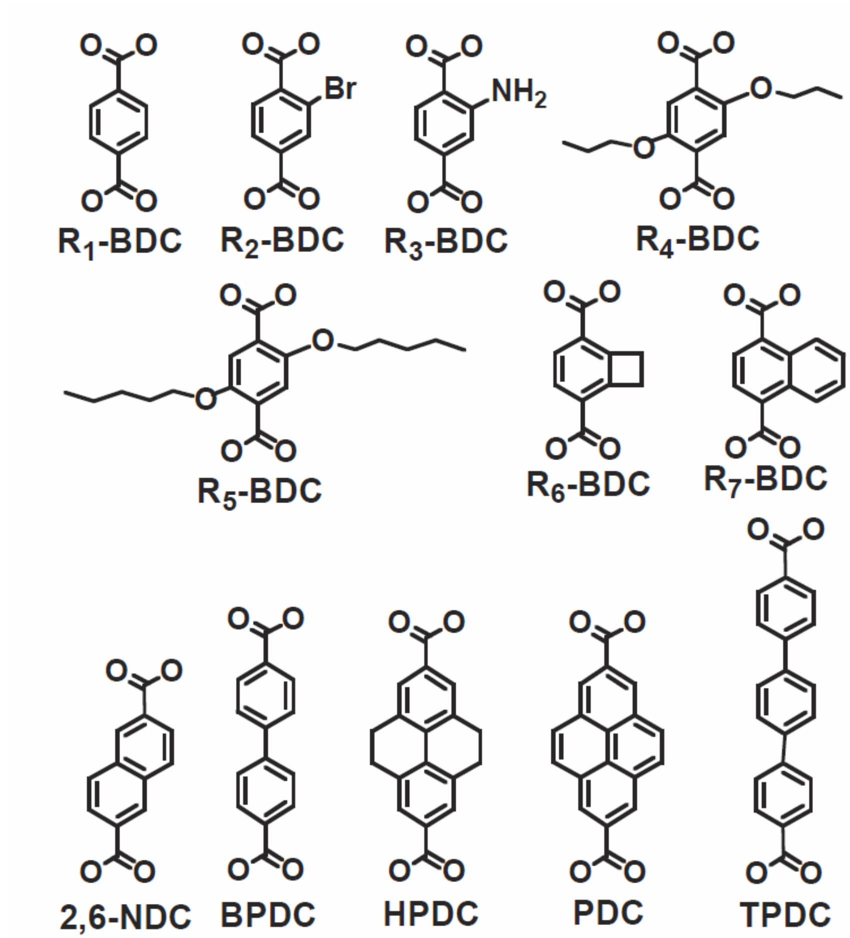
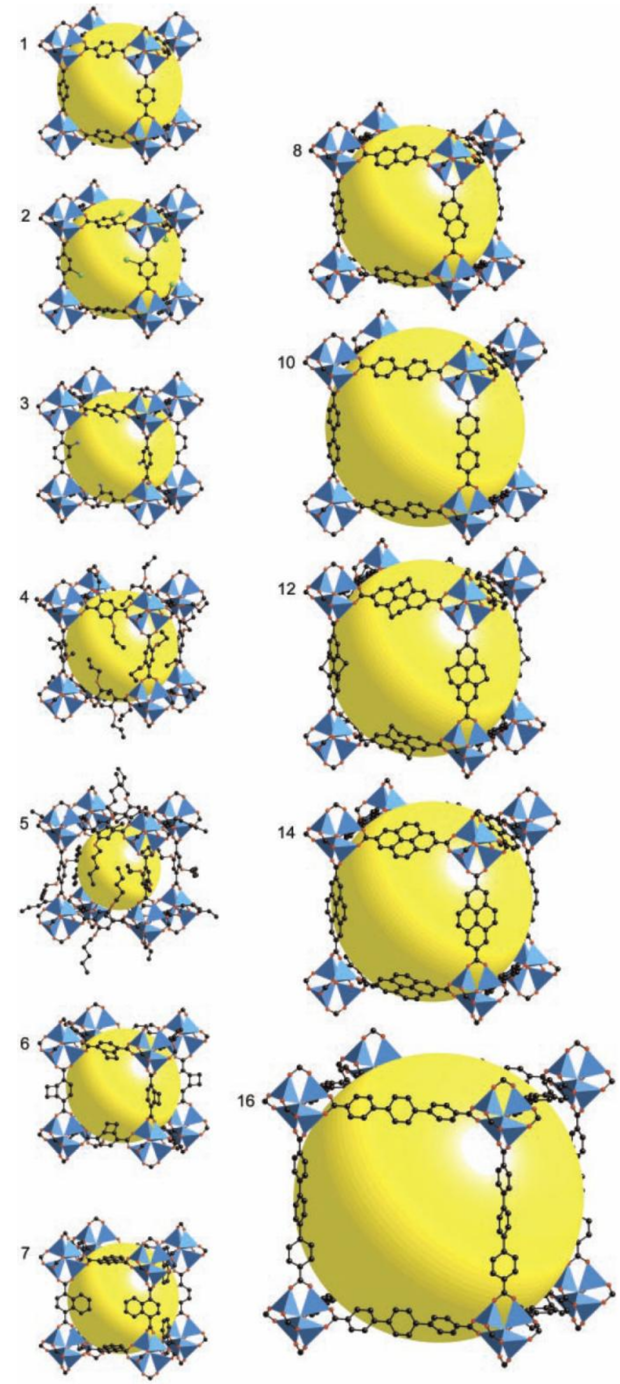
The concept of **isoreticular chemistry** was made popular in **2002** for a series of Zn dicarboxylates but has also been extended to other materials.

Isoreticular MOFs, IRMOFs



(IRMOF-1 = MOF-5)

M. Eddaoudi, J. Kim, N. Rosi, D. Vodak, J. Wachter, M. O`Keeffe and O. M. Yaghi, *Science*, **2002**, 295, 469-472.



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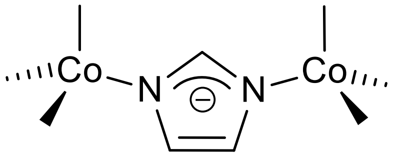
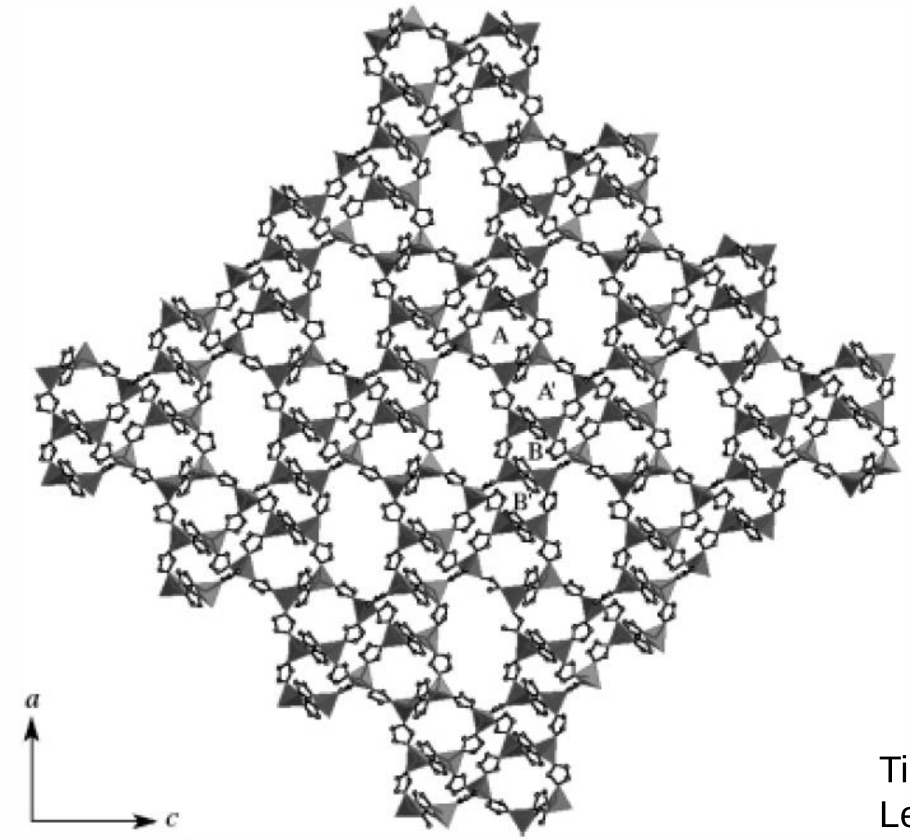
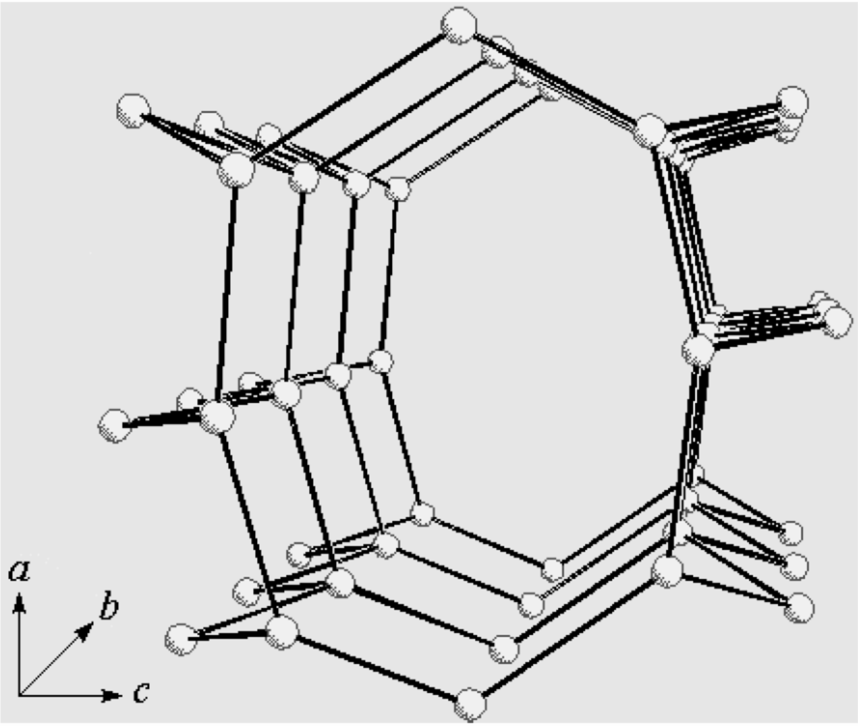
In **2002** (more prominently noted in 2006), imidazolate-based compounds, nowadays known as zeolitic imidazolate frameworks (**ZIFs**), were prepared.

ZIF = zeolitic imid**azolate** framework
$[Co_5(im)_{10} \cdot 2MB]$

MB = 3-methyl-1-butanol

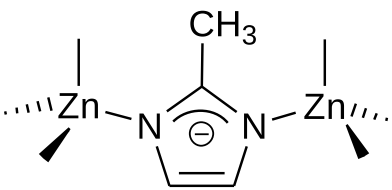


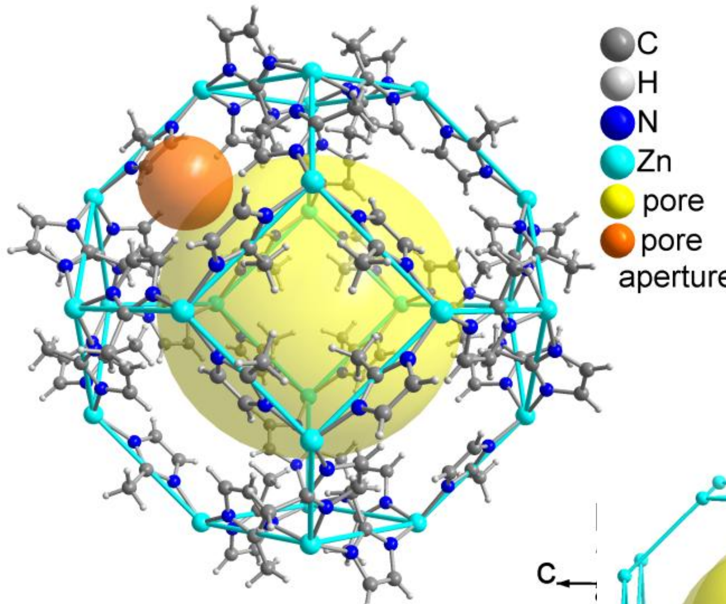




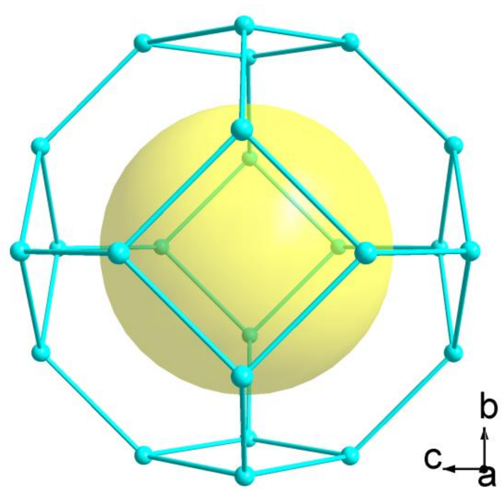
Tian, Y.-Q.; Cai, C.-X.; Ji, Y.; You, X.-Z.; Peng, S.-M.; Lee, G.-H. *Angew. Chem. Int. Ed.* **2002**, *41*, 1384-1386..

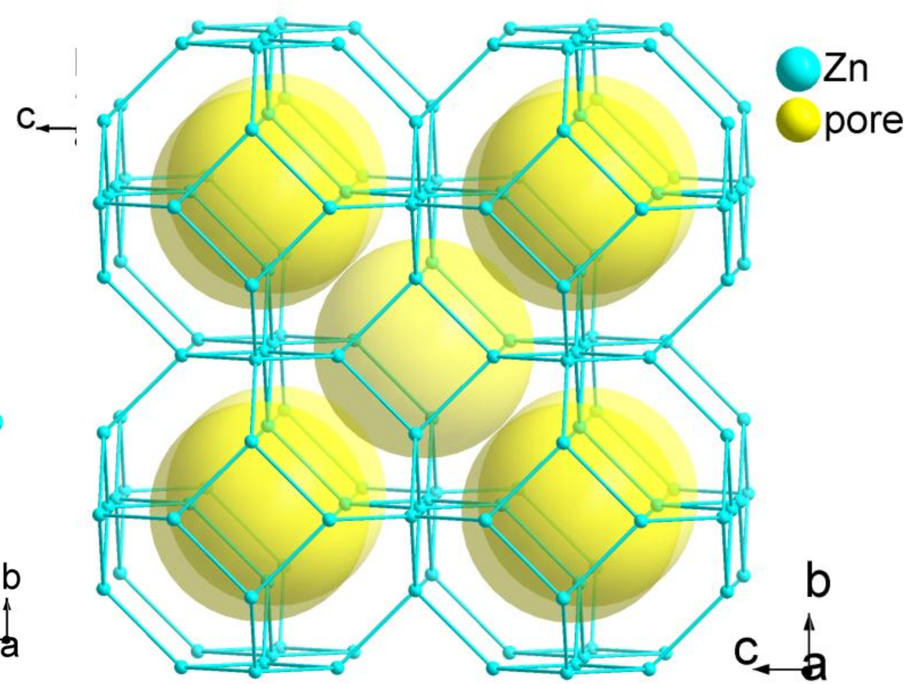
ZIF = zeolitic imid**azolate** framework
ZIF-8 = 3D-[Zn-(mim)$_2$] ·2H_2O





β-cage





Huang, X.-C.; Lin, Y.-Y.; Zhang, J.-P.; Chen, X.-M. *Angew. Chem. Int. Ed.* **2006**, *45*, 1557 –1559.

Park, K. S.; Ni, Z.; Cote, A. P.; Choi, J. Y.; Huang, R.; Uribe-Romo, F. J.; Chae, H. K.; O'Keeffe, M.; Yaghi, O. M. *Proc. Natl. Acad. Sci. U.S.A.* **2006**, *103*, 10186−10191

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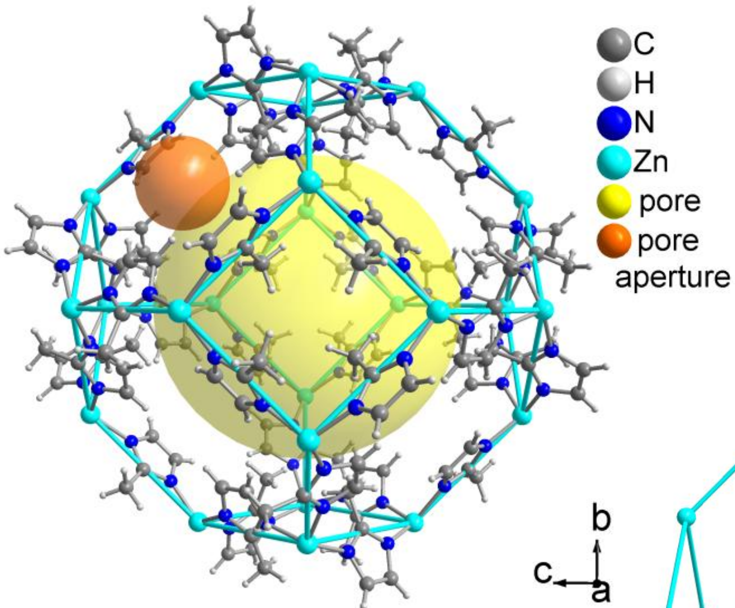
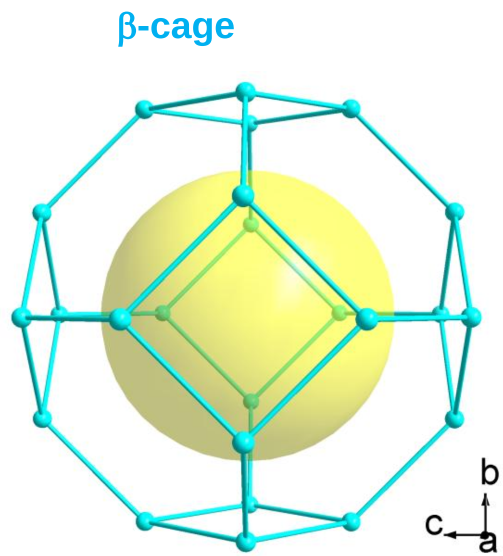
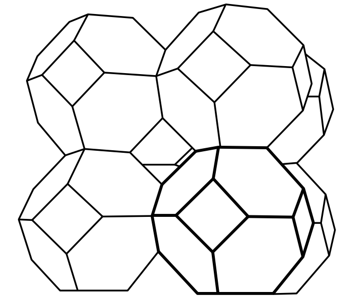
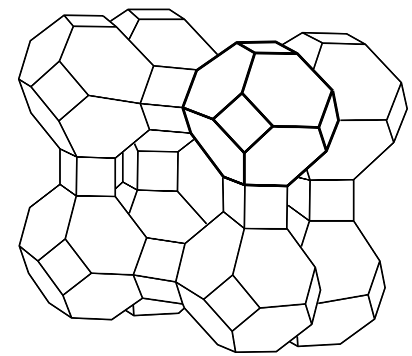
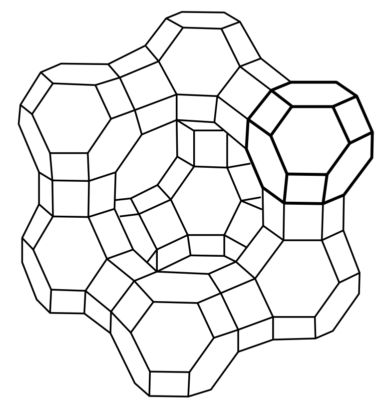


natural zeolites:

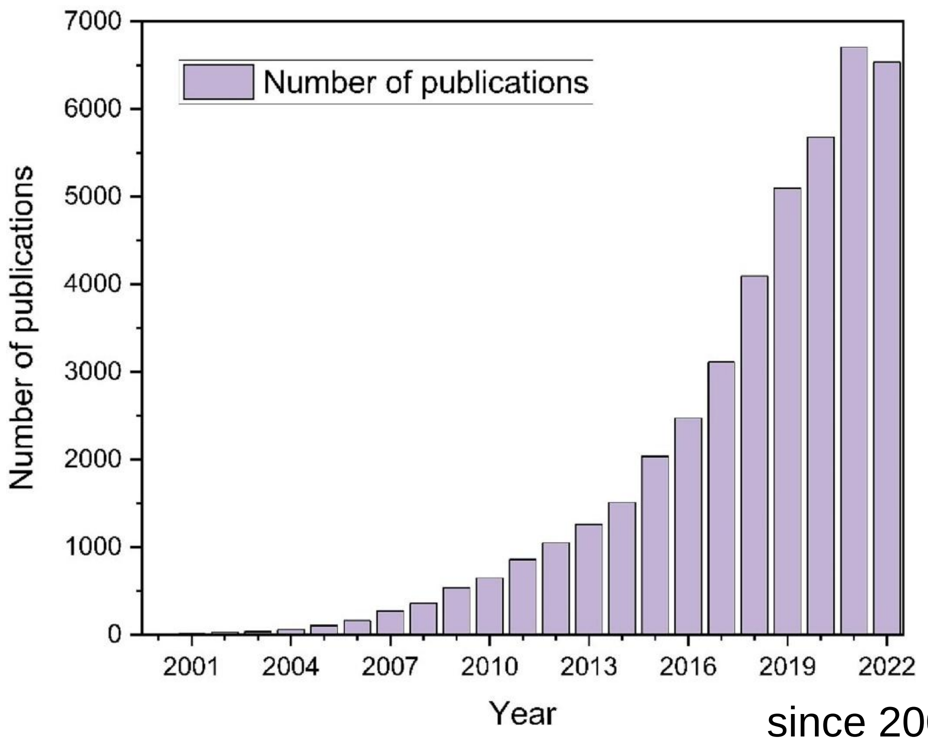








Metal-organic frameworks (MOFs):
A dynamic field



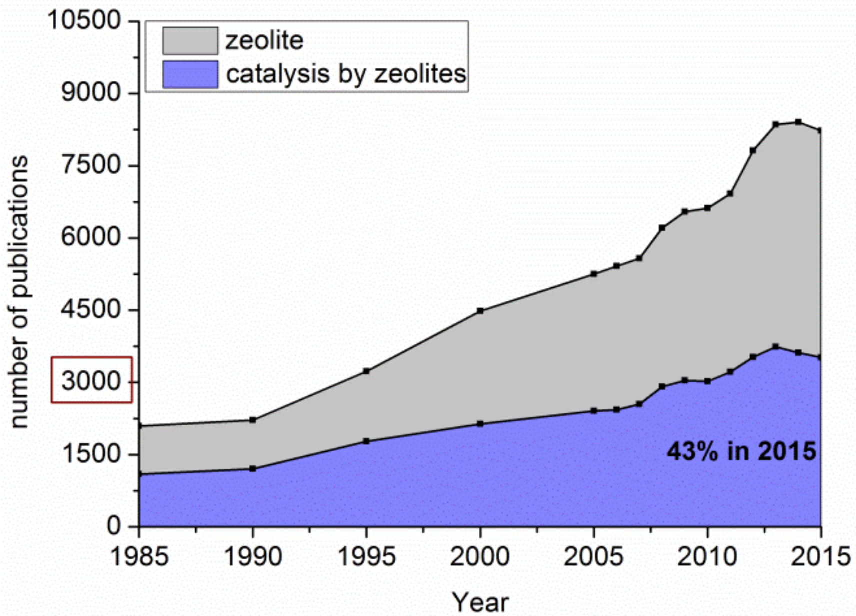
since 2002 the field of MOFs developed with increasing pace ...

... but compared to zeolites ...

P. Wisniewska et al., Chem. Eng. J. **2023**, 471, 144400. https://doi.org/10.1016/j.cej.2023.144400

Metal-organic frameworks (MOFs):
A dynamic field

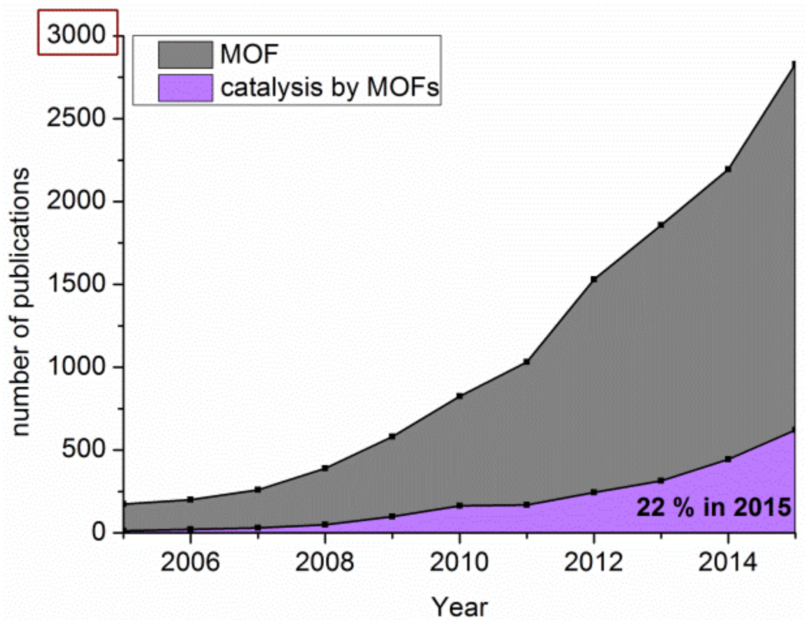
... but compared to zeolites ...



Scifinder " zeolites", "catalysis by zeolites"

MOFs still have to catch up!

Scifinder: "MOF", "catalysis by MOFs"



A. Herbst, C. Janiak,
CrystEngComm **2017**, *19*, 4092-4117
DOI: 10.1039/C6CE01782G

Content

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2. Comparison among porous materials
3. Brief history – the first MOFs
4. Prototypical MOFs – a closer look - new file
5. Synthesis and analysis
6. Selected applications